

Safety and Security of Satellite Communications: A Comprehensive Review of Radio-Frequency and Free-Space Optical Systems

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Abstract As satellite communication (SatCom) systems expand, ensuring their safety and security becomes critical to protect against both unintentional hazards and intentional threats. We first provide an overview of radio-frequency (RF) and free-space optical (FSO) SatComs, comparing their characteristics, applications, and challenges. Subsequently, we clarify the concepts of safety and security, proposing a hierarchical taxonomy of safety and security issues specific to RF and FSO SatComs, along with their respective mitigation and countermeasures. While FSO communication is often perceived as inherently secure due to its narrow, highly directional beams, this perception has led to relatively less focus on its security aspects. Following a systematic taxonomy, we identify significant gaps in FSO security research, highlighting the need for focused investigation in this area. Finally, drawing lessons from the development of RF SatCom systems, we suggest future directions for improving the safety and security of FSO SatCom systems.

Keywords satellite communication, security, free-space optical communication, eavesdropping, quantum key distribution

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1 Introduction

Satellite communication (SatCom) systems are undergoing a profound transformation driven by the convergence of the global broadband demand and the evolution toward beyond-5G and 6G networks[1, 2]. Traditionally, SatCom relied on a relatively small number of geostationary Earth orbit (GEO) satellites providing broadcast and backhaul services using radio-frequency (RF) links in microwave bands[3]. In recent years, however, the paradigm has shifted toward large low Earth orbit (LEO) constellations consisting of hundreds to thousands of satellites with optical inter-satellite links (ISLs), enabling low-latency, high-throughput connectivity on a global scale[4]. Commercial mega constellation projects like Starlink are designed to deliver broadband Internet access directly to end users and to complement terrestrial infrastructure in coverage-limited regions, with significant projected market growth over the coming decade[5]. At the same time, the massive increase in satellites and the emergence of dense space-based networks are reshaping how SatCom is designed, operated, and integrated with other communication systems[6].

As SatCom systems increase in complexity, ensuring their safety and security becomes critical. While RF SatComs are mature and widely deployed, they face significant challenges such as spectrum scarcity, signal interference, and a large attack surface due to their broadcast nature, especially in dense

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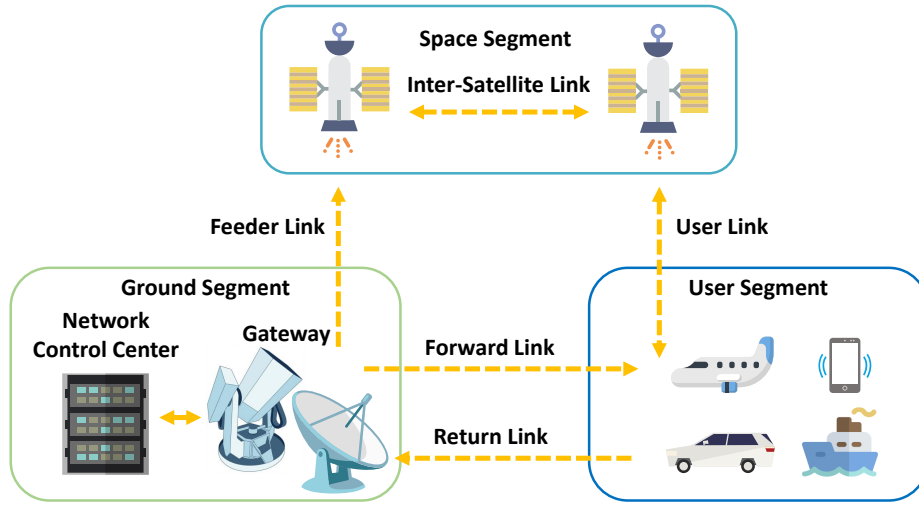


Figure 1. System Architecture of SatComs

constellations[3]. Free-space optical (FSO) SatComs, with its high bandwidth and narrow, directional beams, presents a promising alternative, particularly for inter-satellite links in the space segment[7, 8]. However, both RF and FSO systems are susceptible to unique safety and security risks, ranging from unintentional issues like signal interference and beam misalignment to intentional threats such as jamming and eavesdropping[9–11]. Safeguarding the safety and security of these systems is essential to ensuring the reliability and resilience of SatCom networks as they contend with growing global demand and increasingly sophisticated threats.

1.1 System Architecture of SatComs

A key technological axis in SatComs is the coexistence and interplay between RF and FSO communications. RF links remain the workhorse for many SatCom applications due to their maturity, well-established standards, and robust performance under diverse propagation conditions[3]. However, RF SatComs face fundamental constraints, including spectrum scarcity, regulatory limitations, wide beam footprints, and interference management challenges in dense constellations[12, 9]. FSO SatComs, by contrast, exploit optical carriers in the near-infrared band to provide orders-of-magnitude higher bandwidth with highly directional, narrow beams[7]. This makes FSO especially attractive for high-capacity ISLs, where recent demonstrations and deployments have shown multi-Gbps optical connectivity[7].

Satellite orbit configurations are critical for determining the coverage, latency, and performance of the SatCom systems. The three primary orbit types are low Earth orbit (LEO), medium Earth orbit (MEO), and geostationary orbit (GEO)[3]. LEO satellites, positioned between 500 and 900 km above Earth, provide lower latency and higher bandwidth but require more satellites for continuous global coverage due to their rapid movement relative to Earth’s surface[4]. MEO satellites, orbiting at altitudes ranging from 5000 to 25000 km, offer a balance between coverage and latency, typically used for navigation systems like Global Navigation Satellite System (GNSS)[13]. GEO satellites, located around 36000 km above the equator, can cover a vast portion of Earth’s surface with a single satellite, but suffer from higher latency and signal attenuation due to greater distance. The choice of orbit depends on service requirements, with factors like latency, coverage, and the number of satellites influencing the design of satellite constellations.

SatCom systems are typically structured into three fundamental segments: space segment, ground segment, and user segment. Figure 1 illustrates a typical SatCom architecture. Space segment comprises the satellites and, if present, ISLs. In modern LEO constellations, satellites may be connected by ISLs to form a mesh network that supports satellite-to-satellite communication. If ISLs are not implemented, then connectivity must rely on ground stations, which imposes design constraints on ground station density

and visibility coverage[8]. Ground segment includes gateway ground stations responsible for feeder links, and network control centers that manage satellite operations, resource allocation, and interfacing with terrestrial communication systems[3]. User segment is composed of user terminals—e.g., user handsets, shipborne or airborne terminals, Internet-of-Things (IoT) devices—accessing the network via user links. Starlink provides direct communication between user terminals and satellites, mitigating the dependency on proximal gateways through inter-satellite routing.[5].

1.2 Motivation

As SatCom systems become increasingly complex, a critical issue arises in distinguishing between safety and security concerns. These two concepts are often confused or conflated in the existing literature, leading to fragmented analyses that fail to fully address the different types of risks that SatCom systems face. Safety issues typically involve risks that degrade system performance or cause unintentional disruptions, such as signal interference or misalignment[4, 19], while security concerns focus on intentional threats like jamming, eavesdropping, and spoofing attacks[9, 20]. However, some studies tended to mix these concerns or overlook safety challenges in the broader context of SatCom security, leaving significant gaps in understanding the full range of potential risks and mitigation strategies[8]. This confusion between safety and security needs to be clarified to better design and protect next-generation SatCom systems.

Compared to the extensive research on the security of RF SatComs, the security of FSO SatComs remains underexplored[17, 18]. While FSO is often perceived as inherently more secure due to its narrow, directional beams[15], it faces unique vulnerabilities that have not been adequately addressed in the literature. Much of the existing research on FSO focuses primarily on safety aspects, such as the impact of atmospheric conditions and beam misalignment, while security concerns, including potential optical jamming and eavesdropping, are often overlooked[21, 16]. This imbalance in research focus highlights the need for a comprehensive study of FSO security to understand its potential vulnerabilities and to develop effective countermeasures that complement the security strategies established for RF systems.

1.3 Contributions

The main contributions of this paper are:

1. **Clarifying the distinction between safety and security in SatComs.** We develop a clear conceptual distinction between safety (unintentional hazards, reliability, and availability risks) and security (intentional, adversarial threats) in the context of SatCom systems. Building on this distinction, we propose a unified perspective that links safety and security concerns to specific layers in SatCom. We further organize existing security research into a multi-layer hierarchical taxonomy that spans the physical layer, protocol-level, and trust root, enabling a structured analysis of how different threats and defenses interact across the SatCom architecture.
2. **Highlighting underexplored security aspects in FSO SatComs.** We identify that the security of FSO SatComs remains significantly underexplored compared to RF SatComs. While FSO is often perceived as inherently secure due to its high directionality, latent vulnerabilities remain largely unaddressed, particularly in the physical layer and trust root. This paper underscores the necessity for comprehensive research into the unique security challenges of FSO SatComs.
3. **Lessons from RF SatComs and future directions for FSO SatComs.** By synthesizing decades of experience and research on RF SatCom safety and security, we extract key lessons and map them onto the emerging FSO landscape. Based on this comparison, we identify critical research gaps for FSO SatCom systems and outline promising future directions for enhancing the safety and security of next-generation SatComs.

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews RF/FSO SatCom technologies and surveys existing work on SatCom safety and security. Section 3 introduces the concepts of safety

and security in a general systems context, and develops the unified hierarchical taxonomy to provide a structured survey of safety and security issues and countermeasures across different layers. Section 4 synthesizes lessons learned from RF SatComs, and outlines future directions for FSO SatComs. Finally, Section 5 concludes the paper.

2 Background

In SatCom systems, two primary transmission technologies are widely employed: radio-frequency (RF) communication and free-space optical (FSO) communication. Each technology plays a unique role and comes with its own strengths and limitations.

This section provides an overview of the technical aspects of RF and FSO SatComs, as summarized in Table 1. We then review key research in the areas of RF and FSO SatCom safety and security, highlighting the major contributions in these fields. Finally, we discuss the limitations of previous research.

Table 1. Characteristic Comparison of RF and FSO SatComs

Characteristic	RF SatComs	FSO SatComs
Wavelength	Longer, 4mm ~ 30cm	Shorter, 700nm ~ 1600nm
Bandwidth	Limited with spectrum scarcity	Higher with unlicensed spectrum
Beam Divergence	Wide, less directional	Narrow, highly directional
Alignment Requirements	Tolerant of pointing errors	Sensitive to alignment
Environmental Sensitivity	Less sensitive to weather conditions	Highly sensitive to atmosphere

2.1 Overview of Radio-Frequency SatComs

RF SatComs represent the traditional and most widely used approach in current SatCom systems. The nodes are equipped with RF transceivers and antennas, with information typically modulated onto radio-frequency electromagnetic waves within allocated bands[3]. RF SatComs are highly standardized and well-established, playing a critical role in global communication, from broadband internet services to military and emergency communications[4]. Regulatory bodies define and allocate frequency bands for these communications, ensuring efficient use of the spectrum and minimizing interference[12]. Despite being a mature technology, RF SatComs face several challenges due to the limited and congested RF spectrum[22, 9].

The physical characteristics of RF SatComs are defined by the properties of the electromagnetic waves used for transmission. RF systems operate across various frequency bands, including L, S, C, Ku, K, Ka, and Q/V bands, each with wavelengths ranging from centimeters to millimeters[3]. The beamwidth of an RF system's antenna is an important factor in determining the system's coverage and performance. Half-power beamwidth (HPBW) is an angular width in degrees, measured on the major lobe of an antenna radiation pattern at half-power points[23]. For aperture antennas, the beamwidth of the radiated beam is roughly proportional to the ratio of the wavelength λ to the physical aperture diameter D . In simplified form,

$$\text{HPBW} = k \frac{\lambda}{D} \quad (1)$$

where k is a constant depending on aperture geometry and illumination taper. This means that for a given aperture size, a longer wavelength (typical of RF SatComs) results in a broader beam and larger coverage area. While wide beams allow for greater geographic coverage and robust link availability, they also lead to increased susceptibility to interference, as the transmitted signal can be easily intercepted or affected by unintended recipients within the beam's footprint[22].

Despite their widespread use, RF SatComs face several limitations. The finite RF spectrum, combined with the growing demand for SatCom services, makes it increasingly difficult to achieve high throughput. Regulatory constraints further exacerbate this challenge, as the available frequency bands become

more congested[24]. For instance, RF systems experience interference from other satellites and terrestrial networks. These interference risks are particularly concerning in LEO constellations, where dense deployments of satellites may lead to co-channel interference (CCI), which degrades system performance[12]. Furthermore, the physical size, weight, and power (SWaP) requirements of RF antennas and amplifiers present challenges in terms of satellite design, especially for high-data-rate communications. As a result, RF SatComs must continually innovate in areas like resource allocation, dynamic frequency management, and interference mitigation to remain effective in an increasingly congested and competitive spectrum.

RF SatComs are also vulnerable to a range of security threats, particularly due to their wide coverage and reliance on electromagnetic waves for communication. The broadcast nature of RF signals makes them susceptible to interception, eavesdropping, and jamming by unauthorized actors[8]. Adversaries can exploit the relatively long wavelength and broad beamwidth to interfere with signals, disrupting communications or accessing sensitive data. Additionally, RF SatComs are at risk of spoofing attacks, where an adversary mimics a legitimate satellite or ground station to deceive the system[20]. To mitigate these risks, ongoing research focuses on enhancing physical layer security through advanced encryption techniques, anti-jamming mechanisms, and robust authentication protocols[14]. However, the challenge remains to secure RF communication against evolving and sophisticated threats, particularly in the context of increasing satellite constellations and the potential for adversarial interference[25].

2.2 Overview of Free-Space Optical SatComs

As an alternative to RF, free-space optical (FSO) communication—using laser/optical links—has gained increasing attention for SatComs, especially in ISLs[26]. FSO transmits data via collimated laser beams through atmosphere or vacuum. Because the carrier frequency is many orders of magnitude higher than RF, FSO supports vastly greater bandwidth, enabling high data rates, high capacity, and narrow-beam highly directional links. FSO may achieve data rates tens of times higher than RF and supports lower SWaP (smaller, lighter optics compared to RF antennas), making it attractive for next-generation SatCom systems[15]. Using FSO for ISLs can drastically increase inter-satellite throughput, reduce latency, and improve the efficiency of global mesh connectivity.

The narrow beam property offers several key advantages for FSO SatCom. The shorter wavelength relative to the aperture size allows the emitted beam to be collimated and highly directional, resulting in very narrow divergence angles. Indeed, for FSO systems, a beam divergence angle on the order of micro- or milli-radians is common. Some definitions consider beams with divergence $\leq 0.0057^\circ$ as narrow beams[27]. First, spatial confinement of the signal greatly reduces the probability that an unintended receiver will lie within the beam—enhancing confidentiality and making eavesdropping or wide-area interception much harder compared to RF[11]. Second, because the beam directs energy tightly to the receiver, link budgets can yield low power consumption and efficient throughput, which is especially beneficial for high-capacity ISLs[26].

However, FSO also brings new challenges. The extremely narrow divergence demands very precise pointing, acquisition and tracking (PAT)[15]. Even small misalignment caused by satellite attitude jitter, platform vibrations, or tracking errors can shift the beam off-target, leading to link loss or drastic signal attenuation. Moreover, for satellite-to-ground links, the optical beam must traverse the Earth's atmosphere where environmental disturbances can severely degrade or even break the link. Atmospheric turbulence introduces random fluctuations in received irradiance; scattering or absorption may cause large attenuation; background light can add noise[28]. Finally, line-of-sight (LoS) obstructions such as clouds, buildings, or terrain can block the optical path entirely, making FSO unsuitable for certain scenarios without adaptive solutions like hybrid RF/FSO systems[21].

2.3 Existing Literature

A growing body of survey literature has examined various facets of SatCom safety and security. Below we categorize and summarize these papers according to their emphasis on RF and FSO techniques. Then we critically examine the limitations of existing literature to identify the remaining gaps in SatCom safety and security research. Table 2 compares our paper with existing related survey papers.

Table 2. Comparison with Related Surveys

Ref.	Year	RF SatComs				FSO SatComs			
		Safety	Security			Safety	Security		
			PLS	Protocol.	TRS		PLS	Protocol.	TRS
[29]	2019						✓		
[21]	2021					✓	✓		
[14]	2022		✓	✓				✓	
[30]	2022			✓					
[13]	2022			✓	✓				
[4]	2023	✓	✓	✓		✓	✓		
[31]	2023	✓	✓						
[8]	2024		✓	✓	✓				
[16]	2024					✓	✓	✓	
[17]	2025		✓				✓		
[12]	2025	✓							
[18]	2025		✓	✓					
[32]	2025							✓	
Our paper	2025	✓	✓	✓	✓	✓	✓	✓	✓

Note: PLS, physical layer security; Protocol., protocol-level security; TRS, trust root security.

Several surveys have addressed the security and reliability of RF SatComs, with a focus on various security layers. Yue et al.[4] provided a comprehensive survey of the unique security and reliability challenges in LEO SatCom systems, catalogs known vulnerabilities and risks, and reviews a broad range of mitigation solutions and design guidelines while also outlining future research directions. Similarly, Salim et al.[8] offered a systematic survey of cyberattacks, vulnerabilities, and defense strategies across all three segments of SatCom systems—space, ground, and links—providing a unified taxonomy and evaluation of the state of SatCom cybersecurity. Abdelsalam et al.[17] proposed a cross-layer security framework for LEO space information networks by combining physical layer defense with upper layer authentication based on physically unclonable functions to mitigate denial-of-service (DoS), man-in-the-middle (MITM), and spoofing attacks. Additionally, Tedeschi et al.[14] categorized threats in SatComs, reviewed physical layer and cryptographic defenses, and outlined open research challenges for future SatCom system deployments. Focusing on jamming techniques, Pirayesh and Zeng[9] provided a broad, in-depth survey of jamming attacks across wireless networks (WLANs, cellular, CRN, LoRa, vehicular, GPS etc.), cataloged a wide range of anti-jamming strategies, and highlighted open challenges and future research directions for designing jamming-resilient wireless systems.

Due to the inherent characteristics of FSO communication, it is perceived as being more secure compared to RF communications. This perception has led to relatively fewer studies focused specifically on the security aspects of FSO. Much of the research has been directed toward safety concerns, such as atmospheric influences, beam misalignment, and signal degradation. Patil et al.[16] surveyed existing security strategies for FSO communication, covering encryption techniques, quantum key distribution (QKD), and other defenses. They also identified key research gaps and open challenges for future secure FSO deployments. Lopez et al.[33] analyzed the secrecy performance of FSO links under different eavesdropping scenarios, deriving theoretical bounds on secrecy capacity and outage probability. Endo et al.[34] proposed a novel protocol for FSO-based secret key agreement (FSO-SKA) that uses real-time channel state information to post-select received symbols, thereby improving secret key rate compared to prior FSO-SKA schemes. In recent years, quantum communication has become an emerging topic in FSO SatCom. Diamanti et al.[35] reviewed practical challenges in implementing quantum key distribution (QKD) over FSO links, highlighting the potential of QKD for enhancing security in FSO SatComs. Furthermore, Islam et al.[36] evaluated the practical feasibility of QKD in small satellites by applying

finite-size security analysis, showing that representative LEO missions can still generate secure keys under high channel loss and realistic satellite constraints.

Although a substantial number of studies have surveyed aspects of RF and FSO SatComs, there remain important gaps and limitations in the existing literature that hinder a unified understanding of SatCom safety and security. No prior work appears to deliver a truly comprehensive, cross-technology, multi-layer survey that unifies RF/FSO safety and security under a system-wide threat and defense framework.

First, there is a clear lack of a unified safety and security perspective. Many surveys on RF SatCom security tend to conflate safety with security. For instance, while Salim et al.[8] provided a comprehensive overview of cyberattacks and defense strategies, they did not explicitly distinguished safety concerns such as signal degradation and system interference from security issues. This lack of clear distinction often leads to fragmented discussions and incomplete assessments of the risks in SatCom systems. Similarly, while FSO communication is often perceived as inherently more secure than RF communication[15], this perception is not always critically examined in relation to the unique safety challenges posed by atmospheric conditions and beam misalignment[16, 37].

Second, many works fail to provide a clear hierarchical breakdown of security issues and neglect higher-level and system-wide aspects. A comprehensive security framework for SatComs requires an understanding of threats not just at the physical or link layer, but across the entire system—including ground stations, network management, and overall supply chain reliability. For example, while Tedeschi et al.[14] examined physical and cryptographic defenses, they omitted the need for system level strategies that integrate various defense mechanisms across the entire SatCom architecture. Additionally, Abdelsalam et al.[17] proposed a cross-layer security framework, but their focus remained primarily on physical and protocol-level authentication without addressing broader cybersecurity issues or trust root integration.

Finally, there is a lack of a unified work that covers both RF and FSO in the context of SatComs safety and security. While there have been numerous reviews on RF security, such as those by Yue et al.[4] and Salim et al.[8], and others focusing on FSO communication[16], these studies often fail to address both technologies within a single, integrated framework. RF and FSO are increasingly used together in hybrid SatCom systems, yet the security challenges and mitigation techniques associated with combining these technologies remain underexplored. This lack of integration limits the understanding of how RF and FSO security strategies can complement or conflict with each other in a unified SatCom system.

These gaps highlight the need for a holistic, multi-layered approach to SatCom safety and security, which considers both RF and FSO technologies, addresses all layers of the communication system, and provides a clear distinction between safety and security issues.

3 Safety and Security in SatComs

SatCom systems play a critical role in both civilian and military domains, enabling global connectivity, navigation, and data exchange[2, 6]. However, their safe and secure operation faces significant challenges. It is important to distinguish safety issues (unintentional hazards and reliability risks) from security issues (intentional threats and malicious attacks). Both aspects are vital, and conflating them can lead to blind spots in risk management.

In this section, we first introduce the concept of safety and security. Then we provide an overview of safety and security issues in RF and FSO SatComs, along with measures to overcome these challenges. Finally, we provide a comparative analysis of the existing research and identify the research gaps.

3.1 Concepts of Safety and Security

In complex systems such as SatCom system, it is crucial to differentiate between the concepts of safety and security, as these terms represent distinct, yet often intertwined, risk management domains[38]. We introduce these concepts in general terms, explain the reason why it is important to distinguish between them, and outline typical issues and countermeasures associated with each in SatCom system, as shown in Figure 2.

Safety is generally understood as protection from unintended hazards, e.g., accidents, failures, natural phenomena, human error, or any unintentional event that could result in harm to people, equipment,

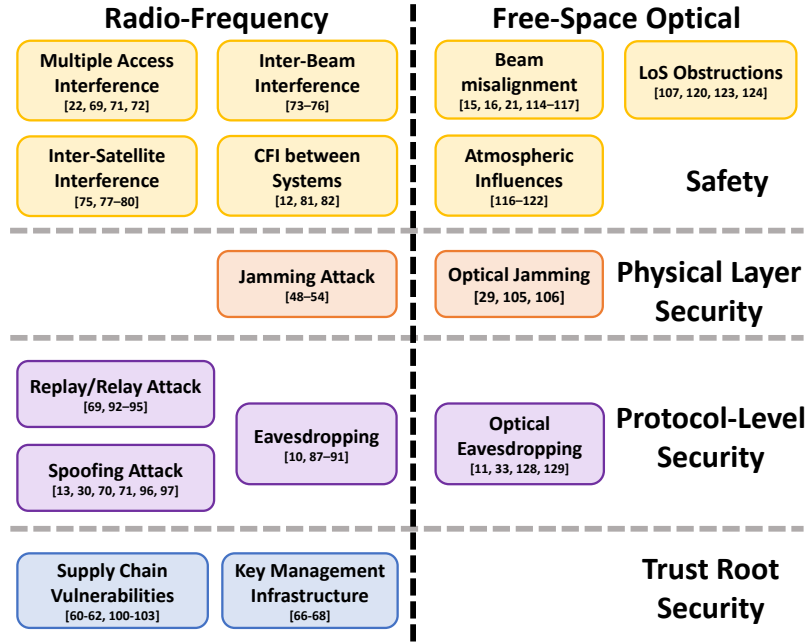


Figure 2. Existing Research of Safety and Security Issues in SatComs

or the environment. Security refers to protection against deliberate, malicious threats like intentional attacks by adversaries seeking to compromise confidentiality, integrity, or availability of systems[39].

The distinction between safety and security becomes important as the two domains often employ different methodologies, standards, and models for risk assessment and mitigation[40]. While safety risks are typically addressed through fault tolerance, redundancy, and error detection, security risks require proactive defense mechanisms such as encryption, authentication, and intrusion detection system.

While safety and security share the common goal of reducing risk and ensuring system reliability, conflating the two can lead to inadequate protection. Safety focuses on preventing unintended hazards, whereas security addresses intentional threats. As such, treating them as interchangeable or overlapping concepts can result in an incomplete understanding of the threats and appropriate mitigations. For instance, while physical layer security in SatComs may deal with both accidental signal degradation and malicious interception, mixing these issues can hinder the development of effective, tailored solutions[41]. Moreover, there are interdependencies between safety and security, as failures in one domain may compromise the other[38]. Conversely, measures aimed at improving safety, such as system redundancy, may inadvertently introduce security vulnerabilities if not carefully designed. Therefore, it is crucial to recognize the distinct but interrelated nature of safety and security in complex systems like SatCom system to ensure comprehensive risk management.

3.2 Radio-Frequency SatComs

Radio-frequency (RF) SatComs remain the backbone of modern satellite networks, providing reliable connectivity across vast geographical areas and supporting a wide range of civilian and military applications[17, 3]. Despite its maturity and widespread deployment, RF SatCom faces a variety of challenges that impact both safety and security[4, 8]. Unlike terrestrial wireless networks, SatComs exhibit unique characteristics, e.g., the long propagation distance, wide coverage footprint, broadcast-oriented transmissions, and limited opportunities for physical access that fundamentally shape their security requirements[4]. As a result, conventional safety and security mechanisms designed for ground networks cannot be directly transplanted into satellite environments without careful adaptation[8]. On the safety side, interference whether arising from intra-system operations, co-channel utilization across systems, or multiple access schemes, can degrade signal quality, reduce system reliability, and in extreme cases lead to service outages[9, 31, 12]. On the security side, RF SatCom is vulnerable to a spectrum of threats,

Table 3. Security Defense and Attack of RF SatComs

Security Layer	Defense	Related Research	Attack
Physical Layer	Frequency hopping	[42, 43]	All attack
	Spread spectrum	[44–47]	
	Time and transform domain filtering	[48–50]	Jamming Attack
	Array antenna	[51–54]	
	Physical layer authentication	[55–57]	Replay/relay attack
Protocol-Level	Authentication and key agreement	[30, 58, 59]	Spoofing attack
			Eavesdropping
			Replay/relay attack
			Spoofing attack
Trust Root	Hardware Security Technologies	[60–62]	Replay/relay attack
	Physical Unclonable Function	[18, 63–65]	Spoofing attack
	Key management infrastructure	[66–68]	

spanning physical layer exploits, protocol-level attacks, and compromise of trust anchors that underpin network authentication and encryption mechanisms[8, 14].

We systematically analyze the safety and security issues associated with RF SatComs. We first discuss the safety challenges, including signal interference within and across SatCom systems. Subsequently, we examine security concerns ranging from physical layer to protocol-level and trust root security.

Safety threats in RF SatComs arise from the fundamental characteristics of satellite links and the increasingly congested space environment[4], as shown in Table 4. With the rapid expansion of LEO mega-constellations and the growing integration of satellites with terrestrial networks, issues such as multiple access interference (MAI) and spectrum scarcity have become prominent. These threats, while not caused by malicious actors, can degrade system performance, reduce communication accessibility, and lead to service outages[22, 69]. Understanding and mitigating these safety challenges is crucial to maintaining robust and reliable SatCom system.

Security threats towards RF satellite systems broadly fall into three categories, illustrated in Table 3. First, at the physical layer, the openness of satellite RF links makes them inherently susceptible to intentional electromagnetic interference, including jamming and distributed denial-of-service (DoS) attacks. These threats can severely degrade signal availability and integrity, and have been extensively studied in the context of GNSS, military satcom, and emerging LEO mega-constellations[31]. The growing use of adaptive waveforms, cognitive radio, and interference-aware beamforming has led to a rich body of anti-jamming and situational-awareness research[70], highlighting the need for resilient physical layer defense mechanisms.

Second, at the protocol level, end-to-end communication procedures face threats such as eavesdropping, and malicious signal injection. The inherently broadcast nature of satellite links enables adversaries to intercept transmissions across wide geographical areas, while the high latency and limited return channels complicate mitigation solutions. As SatComs expand to support IoT and 6G services, secure key distribution, intrusion detection systems, and physical layer security become essential to ensuring confidentiality, integrity, and availability[14].

Third, trust root security has emerged as a critical concern in next-generation satellite architectures. Modern satellite systems rely on distributed trust relationships among transparent/regenerative payloads, ground stations, user terminals, and navigation infrastructures. Compromise or spoofing of any trust anchor—such as falsified satellite identity, forged navigation signals, or malicious downlink overshadowing—can undermine the entire service. Recent studies[4, 13] reveal that LEO constellations and GNSS infrastructures are vulnerable to increasingly sophisticated spoofing and replay attacks, necessitating robust authentication frameworks, physical layer fingerprinting, and secure coordination among network entities.

Table 4. Safety Issues of RF SatComs

Safety Issues	Description	Related Research
Multiple access interference	Interference in shared access multiple-user environments	[22, 69, 71, 72]
Inter-beam interference	Multiple beams of a satellite cause co-channel interference	[73–76]
Inter-satellite interference	Links from different satellites overlap in time and space	[75, 77–80]
CFI between systems	Spectrum sharing in different systems, e.g. LEO SatCom, GEO SatCom, and terrestrial network	[12, 81, 82]

3.2.1 Safety

SatCom systems especially large-scale low Earth orbit (LEO) constellations are inherently exposed to a series of safety challenges arising from physical layer limitations and non-malicious environmental factors[4]. Unlike intentional attacks, these safety issues stem from the fundamental characteristics of satellite links, the harsh outer space environment, and the growing density of both satellite constellations and terrestrial infrastructures.

With the rapid expansion of LEO mega-constellations and the integration of satellites with terrestrial networks in LEO satellite-Integrated Terrestrial Networks (LITNets), spectrum scarcity has become a critical bottleneck[12]. As spectrum is a non-renewable resource, multiple satellite systems and terrestrial terminals are increasingly forced to operate within overlapping frequency bands[83]. This coexistence inherently raises the risk of interference, even in the absence of hostile attackers.

Consequently, modern RF SatComs face several key safety issues. Figure 3 illustrates the four primary types of signal interference issues in RF SatComs. Multiple access interference (MAI) is caused by simultaneous transmissions from multiple users sharing the same communication channel. Inter-beam interference (IBI) arises within the beams of a single satellite due to aggressive frequency reuse or multi-beam payload architectures. Inter-satellite interference (ISI) originates from neighboring satellites within the same or different constellations when their coverage footprints overlap. Co-frequency interference (CFI) between Systems refers to interference between SatComs system and terrestrial communications system.

Multiple Access Interference. Multiple access interference (MAI) is a fundamental performance limiting factor in wireless communication systems, especially in shared access multiple-user environments such as CDMA-based or non-orthogonal multiple-access systems[4]. In satellite downlinks and uplinks where many users concurrently share spectral resources, signals from different users appear as mutual interference at receivers and directly degrade bit-error rate (BER), throughput and energy efficiency. Hence, quantifying and mitigating MAI is central to system design for large-scale LEO constellations.

Recent works[22, 69] adopted a fading-based stochastic model to derive exact expressions for the statistical distribution of MAI and resultant performance metrics under realistic satellite channel models. In particular, Nakagami-m fading is widely used for satellite links because it flexibly models a range of fading severities and fits experimental measurements better than simple Rayleigh/Rician assumptions in many scenarios. These studies derive explicit MAI distributions and exact BER formulas that enable direct evaluation of user-number and reliability tradeoffs.

To control MAI in integrated terrestrial-satellite networks, works[71, 72] on resource allocation propose joint user association, bandwidth assignment and power allocation algorithms that explicitly include cross-tier interference terms in the utility/objective. For example, in uplink NOMA-based terrestrial-satellite architectures Zhang et al.[71] decoupled the original non-convex design into tractable subproblems, employ Taylor approximation and iterative optimization to achieve large gains in system utility while satisfying backhaul and QoS constraints. This class of algorithms directly targets MAI reduction via coordinated resource management across terrestrial and satellite tiers.

Inter-Beam Interference. Inter-beam interference (IBI) arises within the beams of a single satellite when multiple beams reuse the same frequency to increase spectral efficiency[12]. This frequency reuse, while essential for enhancing capacity, inevitably introduces co-channel interference (CCI) in overlapping coverage areas, particularly among adjacent beams operating on identical frequencies. The interference leakage is primarily caused by the side-lobes of the beam radiation patterns and the beam spillover on

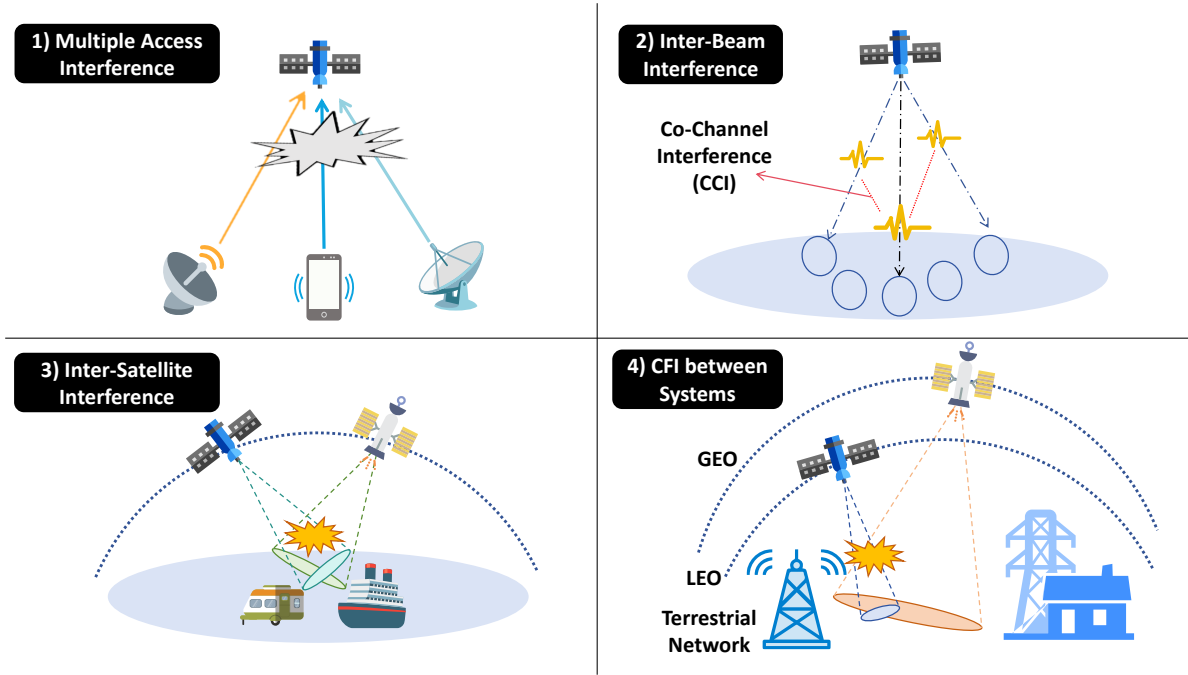


Figure 3. Illustration of Safety Issues in RF SatComs

the ground, which can significantly degrade signal quality and reduce the achievable system throughput. The interference level is commonly quantified using the Carrier-to-Interference Ratio (CIR)[73].

The design of satellite antennas and beam patterns plays a critical role in controlling IBI. Traditional passive antennas generally employ fixed spot beams with uniform beamwidth, which simplifies frequency planning and reduces interference, but may not efficiently accommodate regions with high user demand. Active steerable narrow-beam antennas and phased array technologies allow dynamic beamforming to track high-demand areas, improving spectral efficiency while mitigating IBI[74]. Coordinated interference management schemes, such as the two-color scheme for multi-beam return links, demonstrate that joint MODCOD selection and user scheduling can improve performance by up to 83% compared to conventional four-color schemes, effectively mitigating intra-beam interference [73]. Similarly, joint multi-domain resource-aided interference management (JMDR-IM) schemes for spectrum-sharing satellite-ground networks utilize beam and power coordination to reduce co-frequency interference, achieving higher throughput and lower outage probability[75].

LEO satellite mobility introduces further challenges: as satellites move across heterogeneous regions, coverage requirements and user density vary over time[4]. Fixed frequency reuse patterns may either satisfy hotspot areas but exacerbate interference in low-demand regions, or reduce interference at the expense of capacity in high-demand regions. Hybrid beam schemes, combining wide fixed beams for coverage with high-gain spot beams for traffic hotspots, have been proposed to balance these conflicting requirements[76]. Overall, effective management of IBI requires a combination of adaptive beamforming, dynamic frequency allocation, and interference-aware precoding strategies to ensure reliable and efficient SatComs.

Inter-Satellite Interference. Inter-satellite interference (ISI) arises when links from different LEO satellite systems overlap in time and space, with link distances and angles varying dynamically due to satellite mobility[12]. Modern LEO constellations involve thousands of satellites with interference patterns fluctuating rapidly. Hence systematic mitigation for ISI is required. Two major categories of solutions are commonly used: exclusion zone based methods, joint beam and power optimization. Both aim to maintain reliable coexistence among constellations while respecting regulatory limits on interference, summarized by ITU-R recommendations.

Exclusion-zone approaches focus on controlling ground gateway access strategies using an optimized spatial isolation angle. When gateways of two LEO systems are geographically close and adopt similar

access rules, harmful interference becomes likely. Adjusting the spatial isolation angle can restrict when a gateway is allowed to communicate with a satellite if another system's link is too close in angle. Studies[77] show that optimizing this angle significantly reduces the cumulative interference probability while maintaining system connectivity. Beam disabling or beam hopping at the interfering satellite is used to enforce the exclusion zone dynamically.

Joint beam and power optimization introduces more sophisticated coordination using beamforming, power allocation, and beam-hopping strategies. When coverage overlaps, the achievable Signal-to-Interference-and-Noise Ratio (SINR) depends on time-frequency planning, CSI, traffic load, and beam scheduling[75]. Resource allocation is decomposed into beam-hopping, frequency-band selection, and power allocation, with tools such as water-filling or greedy algorithms used to maximize throughput[78]. Idle co-frequency beams are shut off, and traffic can be rerouted to satellites with interference-free beams via beam switching. This hierarchical optimization ensures that interference to a gateway is minimized without significantly degrading performance of other gateways using the same frequency.

LEO ISI management faces several unique challenges. Dense mega-constellations worsen spectrum coexistence due to multiple operators sharing bands. Residual Doppler differences across user equipments degrade subcarrier orthogonality, causing uplink interference[79]. Dynamic orbital motion leads to time-varying interference environments without reliable statistical knowledge of other systems' allocations. Frequent handovers at high latitudes further increase coordination complexity, making centralized, satellite-based resource management essential to maintain seamless service while controlling ISI[80].

Co-Frequency Interference between Systems. Co-Frequency Interference (CFI) between systems refers to the interference that arises when LEO SatCom systems share spectrum with other systems, such as GEO SatCom systems or terrestrial mobile communication networks[4, 12, 81]. As LEO constellations proliferate and spectrum remains a limited resource, this type of inter-system interference is becoming an increasingly serious challenge.

When LEO and GEO satellite systems operate in the same frequency band, high-level CFI is essentially unavoidable. Because LEO satellites move and frequently cross near the equatorial plane, they can impose increased interference on GEO satellites, especially when the two systems reuse spectrum. According to regulatory frameworks[82], GEO SCSs are often regarded as Primary Users (PU) and LEO systems as Secondary Users (SU), which mandates that LEO systems must limit interference and coordinate to protect GEO operations. This interference coordination is critical. Without proper mechanisms, the dense deployment of LEO constellations will increasingly jeopardize GEO satellite links. As LEO satellites approach GEO's orbital band, the risk of harmful interference rises, particularly in coverage overlap zones. Mitigation strategies may include power control, beam pattern shaping, scheduling, or dynamic spectrum access, but they must be carefully engineered to guarantee that LEO does not cause unacceptable interference to GEO. This coordination will grow even more essential in future 6G-integrated networks, where GEO and LEO jointly contribute to global connectivity.

The second major class of cross-system CFI arises between LEO satellite links and terrestrial mobile communication systems. In envisioned integrated networks, LEO systems may provide coverage in underserved or rural areas, while terrestrial base stations serve denser urban regions[12]. However, when both systems operate in overlapping spectrum bands, especially under spectrum-sharing regulatory regimes, interference from terrestrial BSs to satellite receivers (and vice versa) can be significant. As user equipments choose to connect either to the terrestrial base stations or to a LEO satellite based on signal quality, interference coordination and resource management become essential. Ben Ati et al. [81] provided a comprehensive review of performance metrics in integrated satellite-terrestrial networks, and examines resource allocation frameworks designed to mitigate cross-tier interference through intelligent power control, scheduling, and spectral sharing.

3.2.2 Physical Layer Security

Jamming is the most direct and disruptive physical layer threat faced by RF SatComs. By intentionally transmitting high-power interference in the satellite's operating band, an adversary can severely degrade the received signal quality, reduce the signal-to-noise ratio (SNR), and ultimately prevent the receiver from demodulating or even detecting legitimate transmissions. In many cases, this form of interference constitutes a denial-of-service (DoS) attack, as it aims to sever the communication links between satellites, ground stations, and user terminals[70].

Because most on-orbit satellites, especially transparent satellites, lack sophisticated onboard signal processing, they are particularly vulnerable to power-based/suppression jamming, where the attacker simply overwhelms the legitimate signal with a stronger one. Excessive jamming power can even damage the satellite's receiver front-end. Numerous jamming waveforms exist, ranging from simple broadband noise to deceptive jamming, and can be categorized by the difficulty of generation, required prior knowledge, energy efficiency, and disruptive capability[9].

The inherent characteristics of satellite links enable adversaries to interfere from multiple domains[4]. Space-based jammers, carried by hostile spacecraft, can affect downlinks across wide geographic areas, though with limited dwell time and onboard power. Air-based jammers, deployed on aircraft or airships, offer higher power and greater mobility, enabling burst-type attacks against both uplink and downlink transmissions. Ground-based jammers—often the most powerful and diverse—include fixed, vehicular, or shipborne systems capable of blocking uplink channels and overwhelming satellite transponders regardless of distance.

The operational impact of satellite jamming is profound. Disrupting communication links can delay emergency response efforts, undermine military command-and-control systems, or interrupt commercial services that depend on satellite connectivity[84]. In navigation systems such as GPS, suppression jamming remains a major threat[31]. By saturating the navigation bands with high-power interference, attackers can degrade positioning accuracy or render receivers entirely nonfunctional. Such disruptions may cause aviation and maritime navigation errors, supply-chain delays, and large-scale economic losses. Historically, commercial satellite operators have repeatedly suffered from deliberate jamming incidents, some involving message substitution or sustained interference campaigns that disrupted broadcast services and caused significant financial loss[85].

To mitigate jamming in SatComs, the physical layer can leverage four complementary classes of techniques: frequency hopping, spread spectrum, time-domain and transform-domain filtering, and spatial processing via antenna arrays. These methods exploit orthogonal signal dimensions—frequency, time, and space—to reduce jammer effectiveness, increase required attacker power, or isolate and cancel interference. Together, they form a layered toolbox that can be tuned to diverse threat models, from narrowband or sweeping jammers to wideband, mobile, or spatially concentrated interference sources[41]. Below, we review these major classes of techniques and highlight representative advances tailored for emerging SatCom scenarios.

Frequency Hopping. Frequency hopping (FH) is one of the most established and practically deployed physical layer anti-jamming techniques in SatComs[8]. The key idea is to rapidly switch the carrier frequency of the transmitted signal among a predefined set of hopping slots, thereby reducing the probability that an adversary can consistently jam the active channel. Adversaries must either (i) jam a very wide band, costly in power and detection risk or (ii) track and follow the hop sequence, difficult without prior knowledge. FH variants range from classical fixed hopping to adaptive FH. Each variant trades complexity, spectral efficiency, and robustness to different classes of jammers.

In conventional FH, transmitter and receiver share a hopping pattern and change frequency on a symbol basis. This provides good resilience against narrow-band and partial-band jammers because only a fraction of hops are affected, turning sustained jamming into a higher-cost wideband problem for the attacker. Cognitive Frequency Hopping (CFH) extends this idea by sensing the spectrum and adapting the hopping set and dwell times to avoid busy or jammed channels[42]. CFH therefore combines FH with cognitive sensing for better performance in dynamic interference environments.

Modern analyses of FH variants[43] quantify resilience using metrics beyond simple bit-error rate (BER). A useful metric for CFH is anti-interference tolerance, which captures the probability that a FH link can maintain acceptable performance given sensing errors and convergence delays in the cognitive loop. Closed-form or semi-analytic expressions have been derived to relate sensing performance. Hopping parameters—hop rate, dwell time, hop set size, and jammer strength to end-to-end reliability—providing design rules for parameter selection under different threat models.

Spread Spectrum. Spread spectrum (SS) techniques constitute a well-established physical layer countermeasure against jamming, and are particularly pertinent in SatCom links due to the long propagation distances, relatively low link margins, and high vulnerability to intentional interference. Historically

rooted in military communications for over seven decades, SS is valuable because it spreads a narrow-band information signal over a much wider bandwidth than the original data rate, thus reducing power spectral density and increasing resilience against narrowband or frequency-swept jammers[86].

The fundamental principles of SS and frequency hopping are essentially analogous. When a signal is spread, an adversary that lacks knowledge of the spreading waveform or code must either jam an extremely wide band or attempt to guess the code or sequence. In the context of SatComs, SS can thus both increase availability by mitigating jamming, and improve confidentiality by making the transmitted waveform appear noise-like[44]. Below we summarize the main SS variants and discuss their recent extensions for SatComs.

In direct-sequence spread spectrum (DSSS), the data symbols are multiplied by a high-rate pseudo-noise spreading code such that the resulting signal occupies a much larger bandwidth than the original symbol rate. After transmission and despreading at the receiver, the desired signal regains its original bandwidth and SNR is improved. A survey on security mechanisms with DSSS signals discussed adaptive DSSS schemes balancing jam-resistance and efficiency. A security mechanisms research[45] on DSSS signals studied adaptive DSSS schemes balancing jam-resistance and efficiency. For unintended interference suppression in SatComs, Liu et al.[46] proposed decision-feedback adaptive cancellation on DSSS.

In frequency-hopping spread spectrum (FHSS), the transceiver jumps the carrier frequency according to a pseudo-random hopping sequence shared between transmitter and receiver. The instantaneous dwell time on each hop is long enough to transmit some symbols (slow-hop) or possibly less than symbol period (fast-hop). In FFHSS the hop rate is very high, multiple hops per symbol. In LEO SatComs, FHSS is attractive because the channel may not be constant and hopping adds robustness. For example, Bukhari et al.[47] uses long-range FHSS transceiver designs for LEO support on DtS-IoT (direct-to-satellite IoT).

Multi-carrier direct-sequence spread spectrum (MC-DSSS) is a more advanced variant, which combines DSSS principles across multiple sub-carriers[45]. The data is spread across multiple carriers and each sub-carrier may itself be DSSS-spread, or the system utilises parallel DSSS codes across carriers. This gives additional flexibility in sub-carrier allocation and interference avoidance and can integrate spectrum sensing to dynamically allocate carriers.

Time and Transform Domain Filtering. When the jamming power or coverage exceeds what the FH and SS receiver alone can tolerate, dedicated jamming-suppression filtering techniques in the time and transform domains become important mitigation tools. Time-domain adaptive filtering[48] and transform-domain adaptive filtering[49] exploit the structure of the interference to suppress jammers while preserving the legitimate signal, making them especially relevant for SatCom links with constrained margins and exposed uplink/downlink channels.

In time-domain adaptive filtering, the received baseband signal is passed through an adaptive finite-impulse-response (FIR) filter (or equivalent) whose coefficients are dynamically adjusted (e.g., via LMS or RLS algorithms) to minimise the mean-square error between the filtered output and a desired signal estimate. For example, in satellite navigation and narrowband interference contexts, the work[50] showed how optimal filter length depends on the jammer-to-signal ratio (JSR) and interference bandwidth. Another review on time-domain interference suppression highlighted the trade-off between algorithm convergence, filter complexity and hardware cost in satellite-borne receivers[48]. Time-domain filter can suppress narrowband or sweeping-frequency jammers by creating spectral nulls or adaptively cancelling the interference component prior to despreading or demodulation.

However, time-domain filtering has limitations when the interference is wide-band, rapidly changing, or when the legitimate signal suffers from heavy distortion. For such cases, transform-domain filtering offers advantages that the received signal is transformed into the frequency (or other transform) domain, jammer peaks become more manifest, and adaptive filters operate on individual frequency bins to suppress high-power interference while minimally impacting low-power legitimate signal components. Saulnier et al.[49] demonstrated that transform-domain adaptive filtering in a DSSS context reduced computational load and improved convergence compared to time-domain LMS.

Array Antenna. Anti-jamming techniques based purely on time or transform domain filtering can be limited when the jamming is spatially distributed, multiple jammers exist, or the jamming direction overlaps with the desired satellite link. In these scenarios, spatial resolution through an antenna array becomes a powerful additional degree of freedom. Array antennas used in SatComs exploit the fact that

legitimate satellite signals and interfering jammers generally arrive from different directions. By controlling the amplitude and phase across individual antenna elements, the system can enhance the desired signal while suppressing jammers via null steering, beam-forming, or more advanced processing[70].

Array-based anti-jamming methods encompass a hierarchy of techniques: (i) space-domain adaptive filtering (beamforming/nulling), (ii) space-time adaptive processing (STAP) where temporal taps are added to each spatial channel, (iii) space-frequency adaptive processing (SFAP) where frequency-bins interact with spatial channels, and (iv) joint space-time-frequency processing. For instance, in the satellite navigation and communication literature, STAP has been applied to receiver design to reject both spatially and temporally correlated jamming components[51]. Similarly, SFAP (or space-frequency joint processing) has been proposed for array-antenna anti-jamming under wideband jammers and multi-sub-band interference[52].

One of the key advantages of array-antenna processing in SatCom is increased degrees of freedom (DoF). Consider a multi-element array: by steering nulls towards interfering directions and forming beams towards satellite(s), the system can suppress interference spatially, thereby reducing the effective jammer power at the receiver input without necessarily increasing transmit power or altering the waveform. This spatial filtering is especially beneficial in the satellite context, where uplink/downlink margins are tight and jammers may be highly directional or moving. Sun et al.[53] shows that by adjusting the orientation of the array-antenna on GNSS, residual interference power can be reduced under multiple jammers.

Another important dimension in modern SatCom anti-jamming is joint space-time-frequency processing, which enhances performance when jammers exploit spectral, spatial and temporal diversity. For example, the SF-SED (space-frequency sub-band energy detection) algorithm[54] decomposes array data into sub-bands via an FFT, detects interference-rich sub-bands, and adapts spatial weights per sub-band to improve signal-to-interference-plus-noise ratio (SINR) with reduced complexity compared to full STAP. In SatComs, where wideband jammers and frequency-hopping jammers may appear, such joint-domain processing offers a promising route to robust anti-jamming.

3.2.3 Protocol-Level Security

The protocol-level security in SatComs encompasses the network, transport, and session layers of the Open Systems Interconnection (OSI) model. At this level, the system must address secure routing of signals across nodes, the establishment of sessions and secure transport of data, and management of identities, keys and sessions among heterogeneous terminals and satellites. Unlike purely physical layer threats, protocol-level concerns centre on confidentiality, integrity and availability of communication sessions, proper authentication of entities, secure bootstrapping of trust[14]. In the context of SatComs, protocol design must cope with long propagation delays, constrained terminals, dynamic topology, and broadcast architectures[4]. Figure 4 shows the protocol-level security attacks in the space segment of RF SatComs. We first outline the key security issues in protocol-level: eavesdropping, replay/relay attack, and spoofing attack. Then we discuss the corresponding defense mechanisms for each threat.

Eavesdropping. Eavesdropping remains a primary security concern in SatComs, given the broadcast nature of the transmission and the large coverage areas of satellite beams. In SatComs, particularly in LEO and non-terrestrial networks (NTNs), signals are susceptible to interception by adversarial entities within the footprint of the transmission. The risk of eavesdropping can be exacerbated by the lack of physical barriers in space, where signals propagate over vast distances and are typically broadcast, making it easier for unauthorized receivers to capture and decode the data[8].

Recent research has focused on improving the secrecy performance of SatComs, with a particular emphasis on minimizing the impact of passive eavesdropping. For instance, Jung et al.[10] evaluated the secrecy performance of uplink SatComs under the threat of randomly distributed non-colluding satellite eavesdroppers. By employing beamforming techniques, they derive closed-form expressions for secrecy capacity and outage probability, offering insights for secure constellation design. Similarly, full-duplex satellite-terrestrial downlink security has been investigated using artificial noise and maximal ratio combining, demonstrating that full-duplex operation significantly enhances secrecy compared to half-duplex systems[87].

A secure network architecture for IoT applications has been proposed by integrating inter-satellite beam hopping and optimized relay deployment, effectively mitigating both jamming and eavesdropping

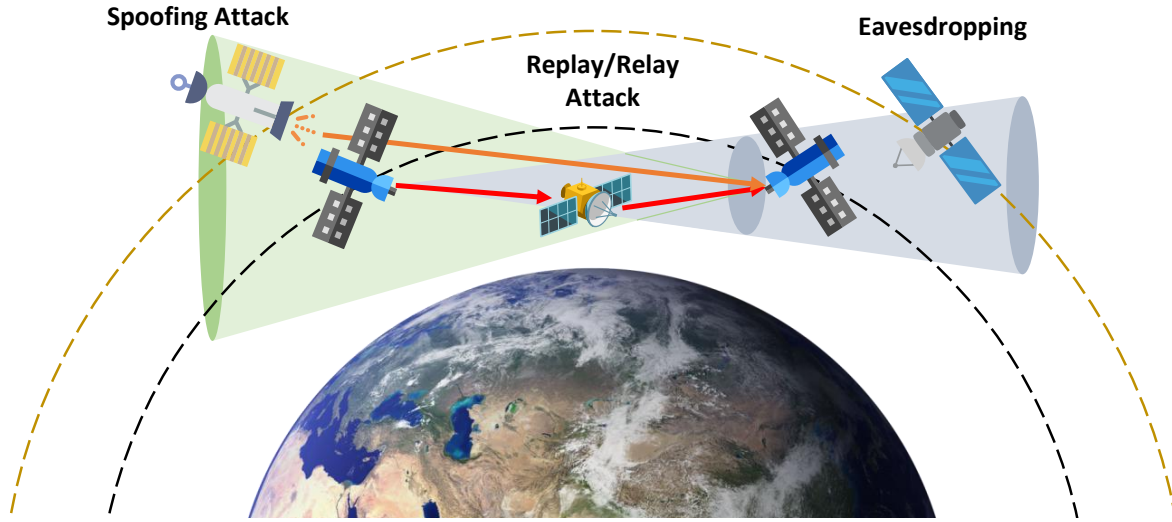


Figure 4. Protocol-level Security Attack in the space segment of RF SatComs

attacks, thereby improving overall system security[88]. Eavesdropping strategies in satellite-terrestrial relay networks have also been analyzed, showing how proactive monitoring by multiple eavesdroppers can be addressed with relay strategies[89].

Robust transmission designs for SatComs under channel uncertainties and eavesdropping threats have been developed using iterative convex optimization, significantly enhancing system security [90]. Additionally, an adaptive power-bandwidth cooperative scheme has been suggested to enhance secrecy in LEO satellite-ground communications against drone eavesdroppers [91].

While physical layer techniques help reduce the probability of an eavesdropper successfully intercepting the signal, strong protocol-level defences is also necessary to ensure that even when an adversary captures the transmission, they cannot derive the original message shared between nodes. Robust key agreement protocols for encryption therefore form a critical line of defence in SatComs. Recent work specifically targeting satellite links has proposed efficient, provably-secure key agreement schemes tailored to the constraints of SatComs. Zhang and Zhai[58] described a temporary identity mechanism used per session to protect anonymity, and the scheme is proved secure under the real-or-random (RoR) model. Another scheme[59] proposed an enhanced authentication with key-agreement scheme for mobile SatComs, exposing stolen-verifier attack weaknesses in earlier work.

Replay/Relay Attack. Replay and relay attacks pose serious threats at the protocol-level in SatComs. Adversaries capture valid signals and later retransmits (replay), or forwards (relay) directly. These actions may bypass conventional cryptographic protections, undermine message freshness, distort protocol state, and lead to outcomes such as session hijack, impersonation, denial of service or incorrect routing[92].

In the domain of satellite navigation and communication systems, several studies illustrate the viability and severity of relay/replay attacks. One set of experiments[92] demonstrated that colluding adversaries can relay real GNSS signals over large distances using off-the-shelf hardware, effectively spoofing the victim receiver despite authenticated navigation messages (OS-NMA) being in place. A related work[93] implemented a distributed and mobile meaconing testbed that uses message-level replay (instead of pure signal-level), enabling reduced bandwidth, multi-constellation replaying, and fine-grained control of victim solutions. These studies highlight that replay/relay attacks exploit timing, location-binding, and signal propagation characteristics rather than purely cryptographic weaknesses. Beyond GNSS, replay/relay threats apply to broader SatComs. Recent surveys[44, 94] of susceptibilities in space information networks noted that transparent satellite payloads can be exploited for relay-based deception, since the satellite may indiscriminately forward signals from unauthorized sources.

Defending against replay/relay attacks in SatComs requires a combination of protocol-level freshness mechanisms, binding between protocol messages and link context, session state uniqueness, authentication integration, cross-layer detection of relayed messages, and careful tuning of these mechanisms to the satellite environment. Defence mechanisms—such as time-stamping, nonce usage, source location validation, and integrated physical layer verification—can be applied in satellite protocol environments[8].

One of the basic and widely-proposed countermeasures is to ensure that each protocol message carries a freshness token—such as a random nonce, a timestamp, or an incremented sequence number—and that the receiver verifies that the token has not been used previously and falls within a valid window. In GNSS-related positioning systems, replay attacks remain effective even if cryptographic authentication is present unless freshness checks are added[95]. Key agreement and session establishment messages should include such freshness elements. Sequence number checks (as in typical IPsec anti-replay windows) are another variant. However, care must be taken in SatComs where large propagation delays and hand-overs may expand the acceptable time window. Hence the window design must reflect the satellite link delays and mobility patterns.

Because relay/replay attacks often introduce additional delay or change the physical characteristics of the signal, protocol defences can tie the message validity to physical context. For instance, in GNSS attack literature[95], Doppler-shift tests are shown to detect replayed/relayed signals by comparing expected Doppler profiles with observed values. In general SatCom protocol design, it may be beneficial to require handshakes or challenge-response messages that are time-bound to expected satellite beam periods, or to embed link co text—satellite ID, beam ID, terminal location or epoch—within the session message so that re-transmissions from another path or after delay become detectable. Often this requires synchronisation between ground terminals and satellite infrastructure.

A successful relay attack often changes physical characteristics even if the protocol appears valid. Therefore, mechanisms that cross-check physical layer metrics with protocol-level validation can help. For example, transmitter fingerprinting can detect replayed signals from a different device. SatIQ[55] uses high-sample-rate radio fingerprinting to detect replay attacks on satellite systems. Incorporating physical layer anomaly detection into protocol validation can enhance detection of relayed or replayed messages.

Spoofing Attack. Spoofing attacks occur when an adversary forges or impersonates a legitimate transmitter at the protocol level (for example a satellite, ground station, or user terminal), thereby deceiving the receiving party into accepting false data, credentials or control messages. In the context of SatComs, the broadcast nature of links, wide coverage footprints, and often limited authentication in legacy down-links make the risk of spoofing particularly acute. Many modern satellite downlink systems can be overshadowed (i.e., spoofed) with modest budgets and hardware, exposing a wide vulnerability[20].

One class of spoofing targets is the Global Navigation Satellite System Spoofing (GNSS) attacks, where false navigation/time signals are broadcast to mislead receivers. Lu et al.[96] showed how spoofers exploit weak downlink signals, broadcasting counterfeit signals to cause erroneous positioning, timing or synchronization. In satellite communications more broadly beyond navigation, the risk extends to control plane spoofing of gateways or user terminals, manipulation of protocol messages, or fraudulent session initiation. Wang et al.[30] categorizes spoofing and authentication attacks in SatCom systems and highlights how spoofing undermines authenticity, integrity and trust in upper-layer protocols.

From a protocol-level perspective, spoofing attacks can take various forms. A ground terminal or adversary pretending to be a legitimate satellite or ground station, injecting session-establishment messages to a user terminal. An adversary forging routing/control messages in an inter-satellite network or satellite-ground link, causing traffic diversion, denial or incorrect routing. In GNSS or satellite tethered networks, adversaries generating fake navigation signals causing the user to mis-locate or mis-synchronise, which then cascades into protocol failures[13]. Errors in expected timing, beam angles, Doppler may be smaller or masked by legitimate dynamics, and protocols with weak identity checks or lacking cross-checking are more vulnerable.

Defending against spoofing attacks requires a multi-layered approach, combining strong identity verification mechanisms with ongoing monitoring and physical layer defences. While traditional upper-layer authentication protocols ensure the legitimacy of data senders, they may not be sufficient against sophisticated spoofers that mimic satellite signals at the radio level. As SatComs landscape continues to evolve,

integrating these spoofing-defence strategies into the design of resilient and secure systems will be crucial to maintaining the integrity of SatComs.

The core of spoofing defence is ensuring that every node can authenticate the identity of the other and bind it to the session or data transfer. Authentication works in SatComs can be classified into five categories: source message authentication (SMA), authentication and key agreement (AKA), handover authentication, cross-domain authentication, and physical layer authentication (PLA)[30]. These authentication protocols reduce spoofing risk by verifying messages originate from valid senders, thus offering integrity and authenticity of sessions.

Because adversaries may spoof at the radio signal level, relying only on upper-layer authentication may not suffice. Novel defences exploit physical layer features of the transmitter or channel to detect impersonation. Convolutional neural networks and autoencoders[56] were deployed to fingerprint LEO-satellite transmitters from I/Q samples and distinguish legitimate from spoofed signals, achieving high classification accuracies using real data from the IRIDIUM constellation. As another complementary approach, FadePrint detects spoofing by fingerprinting the unique fading process of actual satellite links and achieves over 99% accuracy in experiments[57]. Such physical layer authentication measures fortify the defence against spoofers mimicking signal characteristics.

Spoofing may also occur after initial authentication or session takeover, so defences should include ongoing verification rather than one-time checks. Techniques such as transmitter fingerprinting over time, behaviour anomaly detection on message flows, and adaptive re-authentication on handover or unexpected link changes are important. The drone-based aerial spoofing study[97] showed that adversaries using drones at various altitudes could inject spoofed messages into LEO links, and proposed an anomaly detection approach based on autoencoders over physical layer images to reliably detect these sophisticated spoofing attacks. Incorporating such dynamic anomaly detection mechanisms strengthens spoofing resilience.

3.2.4 Trust Root Security

Trust root security refers to the mechanisms, architectures and operational processes by which the foundational, implicit trust anchors of a system are established, managed and protected. A root of trust (RoT) or trust anchor is a component—hardware-based or software-protected—that is implicitly trusted to perform critical security functions (e.g., key management, device identity verification, and certificate validation) and whose compromise would undermine the entire security architecture[98, 99].

We first identify key trust root related security issues specific to SatComs, and then discuss defence and mitigation strategies that are appropriate for the space environment—covering supply chain vulnerabilities, key management infrastructure, and physical unclonable function.

Supply Chain Vulnerabilities. Modern LEO satellites and constellations increasingly rely on commercial off-the-shelf (COTS) components, third-party subsystems, software libraries and outsourced manufacturing to reduce the production cost. This introduces supply chain vulnerabilities: malicious or compromised components might embed root-subversion capabilities, firmware trojans, or allow exfiltration of keys. Recent study[100, 101] demonstrated how COTS components in small satellites can serve as covert persistent channels into onboard systems, bypassing ground monitoring and exploiting implicit trust in hardware items. If a compromised part is integrated into the root-trust chain, mission integrity is undermined.

The supply chain for SatCom involves many tiers of vendors. Thus gaining full transparency into each vendor’s security practices, provenance of components and chain of custody is inherently difficult. Complex supply chains can make it difficult to map which component came from which vendor, which firmware version runs, and whether hardware modifications were introduced. SpyChain[102] showed how COTS auxiliary modules may be inserted, persist in orbit, and subvert operations. Beyond accidental defects, components may be intentionally compromised during manufacturing, test, assembly or distribution. This is particularly acute for COTS components not developed under secure space-grade assurance. Rawlins et al.[103] illustrated how large LEO constellations with unverified components can be hijacked to cause wide-scale disruptions. Many COTS modules incorporate software stacks, open-source libraries or firmware that may contain exploitable vulnerabilities. If these are integrated into satellites without rigorous vetting, the trust root is undermined. The survey[94] on space information networks emphasised COTS software risk.

One of the most effective ways to strengthen trust roots is to anchor them in tamper-resistant hardware such as hardware security modules (HSMs), secure elements (SEs), trusted platform modules (TPMs) or trusted execution environments (TEEs)[60]. These devices generate, store and manage root keys in isolated compartments, preventing direct exposure. SEALSQ[61] described the upcoming Quantum RootKey embedded onboard the satellite to provide a hardware root of trust for post-quantum operations. For a satellite environment, hardware roots include onboard secure modules, tamper-resistant boot systems, key-vaults in ground stations, and authenticated hardware chains of custody. Because satellites cannot be easily repaired once in orbit, monitoring for anomalous behavior is critical. Approaches include telemetry anomaly detection, bus-message profiling, soft-fail safe modes, and frequent health checks[62].

Physical Unclonable Function. Physical Unclonable Functions (PUFs) are hardware security primitives that exploit inherent and uncontrollable physical variations in manufacturing processes to generate unique, device-specific challenge-response behaviors[18]. Because these responses cannot be feasibly cloned or predicted without the physical device, PUFs serve as a ideal hardware fingerprint for authentication, identity binding, and cryptographic key generation without storing keys in memory[63]. PUFs create challenge-response pairs (CRPs) by mapping physical stimuli to measurable responses, leveraging the microstructural randomness of silicon and other materials to establish unclonable identities that resist spoofing and impersonation attack[64].

In the context of SatComs, integrating PUFs into authentication frameworks addresses several core security challenges: limited computational resources on spaceborne terminals, high mobility and frequent handovers. Traditional cryptographic schemes often rely on pre-shared keys or PKI, which are difficult to maintain in the dynamic and resource-constrained satellite environment. PUF protocols, by contrast, derive session keys on the fly from the physical device's unique properties, thus reducing key storage requirements and enhancing resistance to key extraction attacks[18].

Research[65] demonstrated that semiconductor superlattice PUFs (SSL-PUFs) could facilitate secure access authentication and key distribution between satellites and terminals during registration, authentication, and handover phases, ensuring continuity of service even under dynamic topologies. A PUF utilization in satellite-terrestrial integration[63] proposed a novel access and handover authentication scheme for UAV-aided satellite-terrestrial networks enabling 5G. This scheme leveraged PUFs to achieve mutual authentication and key agreement without the need to pre-store secret keys and included mechanisms for physical attack resistance and privacy protection for UAVs and ground terminals. By embedding PUFs in network entities, the protocol avoided vulnerabilities associated with stored secrets and enhances overall resilience to impersonation and replay attacks during handovers.

However, PUFs are not without limitations. Environmental factors such as temperature fluctuations and radiation in space can affect PUF reliability, requiring error-correction mechanisms to ensure stable responses[18]. To overcome this, Chen et al.[64] proposed an SRAM-based PUF architecture that integrates self-healing mechanisms and pre-irradiation masking techniques to improve radiation robustness. Under controlled irradiation up to 100 krad, the proposed PUF achieves a BER reduction by more than two orders of magnitude, demonstrating significant resilience improvements for aerospace applications. Such reliability enhancements are crucial since stable PUF responses directly affect authentication accuracy and key agreement security.

Key Management Infrastructure. The management of cryptographic credentials—generation, distribution, storage, usage, update, revocation and de-commissioning—is a vital component of trust root security in SatCom systems. A robust root-of-trust architecture must not only anchor keys and certificates but also ensure their ongoing lifecycle is securely controlled, auditable and aligned with the constraints of space-borne and ground-segment systems. In SatComs, constraints such as limited physical access, long deployment lifetimes, radiation-hardened hardware, and delayed connectivity impose special challenges. Improper key management—long-lived root keys, inadequate key revocation mechanisms, or insecure key-provisioning of ground terminals—affects trust root security[44, 104].

Classic key management frameworks assume relatively low latency, continuous connectivity with certificate authorities (CAs), and regular maintenance cycles. However, SatComs challenge these assumptions. Building on this, KeySpace[66] framed a set of requirements and evaluations for public key infrastructure (PKI) in highly distributed, delay-tolerant, or intermittently connected space networks. The authors demonstrated that terrestrial PKI techniques can indeed be adapted to space networks if appropriate distribution of authorities and optimized protocols are adopted.

Establishing secure key-generation, distribution, update, revocation and decommissioning processes is essential for trust root security. In satellite systems this involves pre-launch provisioning of root keys, secure satellite commissioning, remote key update capability, revocation of compromised nodes, and careful policy management[67]. Key revocation is perhaps the most challenging stage in space systems. Traditional methods rely on timely links to CAs and low delays, while these assumptions break in SatCom environments. SoK on PKI[68] described how revocation and credential management suffer from orbital dynamics, delay and partial connectivity.

3.3 Free-Space Optical SatComs

Free-Space Optical (FSO) SatComs—also referred to space laser communication or optical wireless satellite networks (OWSNs)—exploit narrow, highly directional optical (laser) beams to transmit data through free space, rather than relying on radio-frequency (RF) carriers[15]. Because optical frequencies lie orders of magnitude above RF, FSO links enjoy a far larger unlicensed spectrum, enabling very high data rates, reduced size/weight/power (SWaP) of transceivers, narrow beam-divergence, and lower probability of interference and interception, compared to conventional RF SatComs. Since line-of-sight (LoS) connectivity is sufficient and fiber or physical RF infrastructure is not required, FSO SatComs offer a promising means to connect remote or isolated areas as long as satellite visibility is maintained. This flexibility enables integral to a variety of applications, including inter-satellite links, satellite-to-ground communication, deep space missions, and satellite-to-airplane communication, offering a potential alternative to traditional RF links[21].

Table 5. Security Defense and Attack of RF SatComs

Security Layer	Defense	Related Research	Attack
Physical Layer	Multiple-input multiple-output	[29, 105, 106]	Optical jamming
Protocol-Level	Authentication and key agreement	[30, 34, 107, 108]	Eavesdropping
	Quantum key distribution	[32, 35, 36, 109–113]	
Trust Root	Limited research available		

Unlike RF systems, which are subject to interference and spectrum congestion, FSO systems can provide highly secure communication channels due to their narrow, directional beams that are less susceptible to interception. However, FSO SatComs also face unique challenges, safety issues including beam misalignment, atmospheric attenuation, LoS Obstructions, and emerging security concerns. Addressing these challenges is crucial for the successful deployment and operation of FSO SatComs. Table 6 summarizes the key safety issues of FSO SatComs along with related research. However, research on the FSO SatCom security remains limited compared to RF SatCom, especially in the context of trust root security, as shown in Table 5.

3.3.1 Safety

Whereas conventional RF SatComs face safety risks such as interference, spectrum congestion, and co-channel conflicts, FSO SatComs confront a different suite of safety challenges rooted in the physical nature of optical propagation, system alignment requirements, and environmental variability[37]. Key safety issues for FSO satellite systems include beam misalignment, atmospheric influences, and LoS Obstructions[16]. While beam misalignment occur in all FSO links, atmospheric influences and LoS obstructions are particularly relevant for links between satellites and ground segment, where the optical beam must traverse the atmosphere of Earth. These safety issues can lead to signal degradation, increased bit-error rates (BER), link outages, and potential data loss, thereby compromising the reliability and performance of FSO SatComs.

Beam Misalignment. FSO satellite links employ extremely narrow laser beams—often only micro-radians in divergence—to achieve high gain, which makes accurate pointing and tracking crucial. Any

Table 6. Safety Issues of FSO SatComs

Safety Issues	Description	Related Research
Beam misalignment	The narrow laser beam deviates from intended path, causing signal loss or degradation	[15, 16, 21, 114–117]
Atmospheric influences	Weather phenomena and atmospheric turbulence may distort or attenuate the optical signal	[116–122]
LoS obstructions	Physical obstacle blocking the path will disrupt the communication	[117, 120, 123, 124]

slight misalignment between the transmitter and receiver line-of-sight can lead to severe signal loss[21]. Satellite platform vibrations, jitter in pointing mechanisms, and imperfect tracking can all cause the beam to wander off the photodetector aperture[16]. The safety risk here is the potential for sudden link drops or degraded signal quality even in clear weather or outer space, simply due to pointing issues. Satellite micro vibrations and pointing jitter is a major performance limiter for inter-satellite optical links[15].

FSO links cannot maintain stable if the terminal is subject to vibrations, as even tiny angular deviations result in tracking misalignment, reduced received power, and high BER[114]. In essence, the narrow optical beam can easily miss its target by a few micro-radians of angular error, causing an outage. The problem is exacerbated for smaller satellites with limited attitude control. They experience systematic pointing biases in addition to random jitter. Modeling studies[115] classify pointing errors statistically. Pure random jitter yields a Rayleigh or Gaussian pointing error distribution, whereas bias combined with jitter leads to Rician or Beckmann distributions. These models have been used to quantify power loss and outage probability due to mispointing. For instance, El-Nayal et al.[116] evaluated optical uplinks to LEO and found that pointing errors at low elevation angles can cause several dB of additional attenuation, greatly diminishing the link margin needed to achieve a given BER. In one scenario, a mild pointing offset combined with light fog required over 25 dB more transmit power to maintain the same BER as a perfectly aligned clear-weather link. Beam misalignment can induce signal fades independent of weather and demands precise control systems to avoid data loss.

A combination of mechanical stabilization, real-time tracking corrections, beam parameter tuning, and system-level redundancy is used to safeguard against beam misalignment. By minimizing pointing error sources and rapidly correcting any deviations, these techniques maintain the optical link's integrity even under satellite vibration and dynamics.

Ensuring robust pointing, acquisition, and tracking (PAT) is vital to mitigate misalignment issues. One straightforward method is to increase the beam divergence so that the beam footprint at the receiver is larger and less sensitive to small pointing errors[114]. This comes at the cost of reduced power density, but it provides a larger safety margin for jitter. Indeed, system designers often trade off some beam tightness to accommodate expected platform vibrations.

Complementary to this, many studies focus on active control schemes to stabilize the beam. Vibration isolation at the transmitter or telescope mount can physically damp high-frequency jitters. Additionally, fast steering mirrors and inertial sensors are used in closed-loop tracking systems to actively compensate for motion. Kadir et al.[114] surveyed various approaches to counteract pointing disturbances: bandwidth adaptation, beamwidth adaptation, power control, adaptive coding, spatial diversity, vibration isolators, and feed-forward tracking compensation. Adaptive power control can temporarily boost the laser output when misalignment loss spikes, and forward error correction can mask brief errors due to pointing fades.

Another crucial mitigation is improving the pointing accuracy of the terminals through better sensors and actuators. Modern optical terminals use star trackers, fine-pointing cameras, and reaction wheels to achieve sub-microradian pointing stability. For instance, a transportable optical ground station featured a telescope with coarse pointing accuracy $\sim 100\mu rad$ and a secondary fine-tracking loop with $\sim 10\mu rad$ precision to keep the beam locked on target[117]. Such PAT systems ensure the beam remains aligned despite satellite motion.

Atmospheric Influence. Free-space optical links between ground stations and satellites are highly susceptible to atmospheric conditions. Weather phenomena like fog, rain, snow, and clouds introduce significant attenuation by absorbing and scattering the laser beam, often degrading or even completely

cutting off the link[118, 119]. For instance, fog is one of the most severe attenuators—dense fog or low-lying clouds can impose tens to hundreds of dB of optical loss per kilometer, effectively blocking the beam and causing link failure[116]. Even moderate fog can be more detrimental than rain or haze, yielding worse signal fade and availability reduction. Cloud cover is particularly problematic: unlike radio waves, optical signals cannot penetrate most clouds, so a thick cloud in LoS will terminate the connection entirely[119]. These atmospheric attenuation factors directly threaten the reliability and safety of FSO SatCom links by increasing the probability of communication outages and data loss during adverse weather.

Atmospheric turbulence is another critical influence. Due to temperature and pressure variations, turbulence arises from random refractive index fluctuations in air and causes the optical beam to scintillate. The received signal intensity fluctuates rapidly, leading to deep fading and high bit-error rate (BER)[16]. In severe daytime turbulence or near the ground, the beam can even wander off the receive aperture, exacerbating misalignment issues. Research[118] has quantified these effects using models for the refractive index structure parameter C_n^2 and Rytov variance to predict scintillation severity. Satellite uplinks tend to suffer stronger turbulence early in the beam's propagation, causing more severe scintillation than downlink. Maharjan et al.[120] showed that an uplink optical beam is distorted near the ground, potentially reducing power at the satellite more than a downlink beam which only encounters turbulence at the end of its path. In one analysis of a 500 km LEO uplink[121], measured weather data from Cairo indicated that link availability would drop precipitously whenever visibility fell below 5 km in heavy fog or haze conditions. Such findings underscore that atmospheric effects are the dominant safety risk for FSO SatCom, often limiting link uptime to clear-weather periods.

To address atmospheric risks, researchers propose a combination of diversity, adaptive techniques, and robust system design. A primary strategy is site diversity, using multiple optical ground stations at different locations so that if one site is clouded out or fogged in, an alternate site with clear skies can take over. By networking many geographically separated ground stations, the system can significantly improve the probability that at least one station has a clear line-of-sight[117]. Studies[123] showed that a distributed network can achieve availability 99.9% even in adverse climate regions by weather hopping between stations. Another approach is adaptive transmission, dynamically adjusting the link to current weather. For example, an experimental system[116] used a visibility sensor to adapt the transmit power under fog or heavy rain, the laser power is increased to maintain the required signal-to-noise ratio (SNR), greatly improving the link's SNR and BER during weather events. Likewise, selecting an optimal wavelength can mitigate some atmospheric effects. Longer wavelengths around 1550 nm suffer less Rayleigh scattering and have been found preferable to 850 nm in foggy conditions[122]. Simulation analyses[120] confirm that 1550 nm is the preferred choice to mitigate turbulence and minimize atmospheric absorption, compared to shorter optical bands. Increasing the receiver aperture size also helps collect more of the distorted beam, reducing losses from beam wander and scintillation. For turbulence distortion specifically, deploying adaptive optics (AO) at the receiver can correct wavefront aberrations in real-time. AO systems can significantly improve coupling of the incoming beam into detectors or single-mode fibers by flattening the phase front, which is essential for coherent optical communication modes[125].

LoS Obstructions.. FSO systems use essentially straight-line laser propagation, hence any physical obstacle blocking the path will disrupt the communication. Line-of-sight (LoS) obstructions refer to anything that geometrically intervenes between the optical transmitter and receiver. In SatComs, LoS obstructions are primarily a concern for the ground segment, especially when the ground station is located in urban or mountainous terrain. Unlike lower-frequency radio waves, which can sometimes diffract or penetrate materials, optical beams cannot bend around obstacles significantly[21]. Thus, if a satellite is near the horizon and a mountain ridge or building lies in between, the optical link will be cut off until the satellite rises above that obstacle. This raises safety concerns in terms of link availability—misalignment can cause brief fades, and weather can be transient, but a hard blockage by terrain or structures absolutely prevents communication. Such occlusions could create predictable communication blackouts in certain directions. Additionally, the very Earth itself becomes an obstruction outside of scheduled visibility windows. LEO satellites are only above the local horizon for short durations, and beyond that the planet blocks the link[120]. This intrinsic LoS limitation means FSO downlinks have duty cycles constrained by orbital geometry and any further obstructions only shorten the available contact time.

Research on LoS obstructions in FSO SatCom tends to focus on link availability analysis and site selection. Link budget studies[123] explicitly treat any periods of cloud cover or geometric blockage as

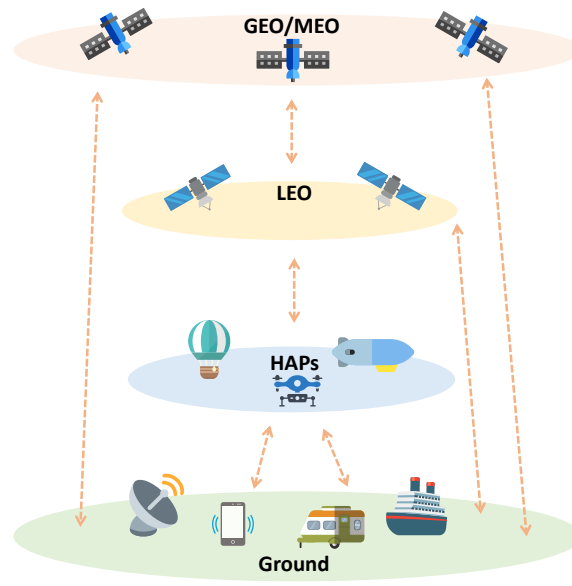


Figure 5. High-Altitude Platform Station (HAPS) Relays in FSO SatComs

link outages in availability calculations. This is an important safety consideration for network design, as missing the first or last few minutes of every pass could mean significant data return loss.

The primary mitigation for LoS obstructions is proper site selection and ground terminal placement. Optical ground stations are ideally situated in locations with clear horizons and minimal surroundings. If an optical station must be in a developed area, installing the terminal on a tall building or tower can elevate it above many local obstructions. Mission planners use tools to compute visibility masks and choose sites with low mask angles so that satellites can be tracked from the moment they rise above the horizon until they set. In scenarios where some obstructions are unavoidable, an effective measure is increasing site diversity with multiple ground stations in different locations[117].

Another approach is to enforce an elevation angle cutoff in operations, only initiating optical links when the satellite is above thereby avoiding most ground clutter and thick atmosphere. Researchers have also explored mobile or deployable ground stations that can be moved to alternate sites if needed[117]. In the long term, high-altitude platform stations (HAPS) acting as optical relay could leap over obstacles and mitigate atmospheric influences[124], as shown in Figure 5. By eliminating obstructions through smart placement and using diversity, FSO networks can greatly reduce the risk of link loss due to LoS issues. Each of these measures improves the overall link safety by ensuring that a clear optical path can be maintained or quickly restored, thus preserving continuous communication whenever possible.

3.3.2 Physical Layer Security

FSO communication offers appealing advantages over RF communication: high data rate, narrow-beam, line-of-sight (LoS), license-free spectrum, and relative immunity to typical RF-based interference or electromagnetic interception[126]. Nonetheless, FSO SatCom links are not immune to physical layer security threats, especially those leveraging the distinctive characteristics of the optical propagation medium.

One of the most prominent security concerns at the physical layer for FSO systems is optical jamming—i.e., adversarial transmission of disruptive optical signals to interfere with the legitimate FSO beam, causing degradation in signal integrity, increased bit-error rate (BER), or complete denial of service (DoS). Despite the high directionality of optical beams, jamming remains feasible in ground segment, especially if the adversary knows or can infer system parameters (wavelength, pointing, receiver FoV) and accesses a vantage point allowing injection of interfering optical signals[127].

Although there are few optical jamming studies in SatComs context, significant amount of research focus on optical jamming in ground segment. These studies may provide valuable insights for addressing

future optical jamming in FSO SatCom systems. Sliti et al.[105] highlighted that FSO systems are vulnerable to jamming on account of limited operational wavelengths, potentially wide receiver fields-of-view (FoV), and because adversaries can more easily identify link geometries in many deployments. A detailed analysis[29] showed that multiple-input multiple-output (MIMO) FSO system can significantly reduce the effect of jamming when spatial diversity is employed. Other works also considered the applications of dynamic resource allocation and reflective surfaces to counter reactive jamming. A recent work[106] proposed a mirror-element-based scheme (RJA-MEA) to dynamically allocate resources and suppress jamming in FSO networks.

3.3.3 Protocol-Level Security

In FSO SatComs, eavesdropping are significant protocol-level security concerns. However, compared with traditional RF communication, the potential for eavesdropping in FSO systems is much smaller due to the inherent characteristics of the optical transmission[8]. The primary advantage of FSO over RF in terms of eavesdropping resistance lies in the narrow, highly directional beams of light used in optical systems. This focused transmission makes it difficult for an unauthorized receiver to intercept signals unless they are positioned directly within the LoS of the optical link. In contrast, RF signals propagate over wider areas, increasing the likelihood of interception by adversaries in the vicinity[4].

However, eavesdropping vulnerabilities remain in FSO SatComs. A eavesdropping model was proposed[33], proving the feasibility of eavesdropping in FSO communications. Eavesdroppers placed close to the legitimate transmitter can easily intercept the communication with the attenuation optical beam, as shown in Figure 6. A spacecraft attacker can also eavesdrop on optical links based on high altitude platform station (HAPS)[11]. Three different scenarios—Inter-HAPS communication, ground-HAPS uplink communication, and HAPS-LEO satellite uplink communication—are evaluated in the paper[128]. The performance of these systems, particularly in terms of secrecy, is degraded by atmospheric turbulence, which can cause fading and signal degradation, making the optical signal more vulnerable to interception.

Furthermore, the growing interest in mixed FSO/RF integrated satellite-HAPS relaying networks presents new challenges. A study[129] on the secrecy performance in the presence of eavesdropping demonstrated that heterodyne detection techniques for FSO are more secure than direct detection under various channel conditions, thus providing a possible countermeasure to eavesdropping in such integrated systems.

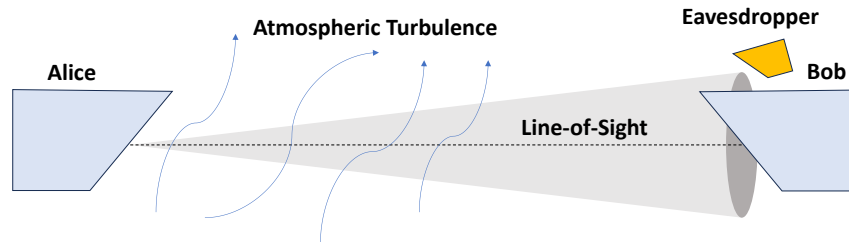


Figure 6. Eavesdropping in FSO Communications

To mitigate the risks of eavesdropping in FSO SatComs, recent studies have focused on two main areas: authentication and key agreement, and quantum key distribution (QKD).

Authentication and Key Agreement. The high directivity and LoS nature of FSO SatComs reduce but do not eliminate protocol-level risks. Authentication ensures that only legitimate nodes establish links and key agreement ensures the confidentiality and integrity of subsequent communication. Limited by the constrained computational resources in SatComs, an efficient and robust authentication and key agreement scheme is critical.

Endo et al.[107] conducted one of the foundational works, in which they performed the first full-field experimental demonstrations over a 7.8 km terrestrial FSO link, achieving symmetric key rates from ~ 100 kbps up to ~ 7.77 Mbps under varying atmospheric and beaming conditions, even when total losses reach 55 dB. Building on that, they further[34] used post-selection based on instantaneous channel state

to discard low-quality symbols, thereby improving the secret key rate. While these are terrestrial FSO experiments, the principles and techniques—channel randomness, post-selection, linking key generation to physical layer observations—are of guiding significance for satellite optical links, especially when combined with authentication to prevent man-in-the-middle, impersonation, or active attacks.

Recent work[108] directly targeted inter-satellite optical links. It proposed a dual-layer protocol combining physical-domain parameters with cryptographic authentication. This design uses a challenge-response framework to authenticate the two endpoints before establishing the link. By tying authentication to physical layer properties of the optical link in addition to cryptographic verification, this method defends against spoofing and impersonation attacks in dynamic multi-satellite constellations.

Pure FSO authentication and key agreement literature is limited, works in SatComs more broadly may be adapted to FSO context[30]. These cryptographic schemes do not inherently rely on physical layer optical channel properties—meaning they are applicable regardless of whether links are RF, FSO, or hybrid—and thus offer a complementary layer of security when used alongside physical layer security.

Quantum Key Distribution. Quantum key distribution (QKD) is a cryptographic method that allows two remote parties to establish a shared random secret key by exchanging quantum states over a quantum channel, plus classical authenticated channel. Compared to classical key exchange, QKD has a distinguishing feature that its security is rooted in fundamental laws of quantum mechanics rather than computational assumptions[109]. Even an adversary with unlimited computational power cannot break the key without being detected. Once the shared key is generated by QKD, it can be used in classical symmetric encryption, exactly as in classical cryptosystems. QKD itself does not transmit message data but only establishes a secure shared key. QKD offers a promising path to quantum-safe key distribution over optical SatCom links.

The launch of Micius in August 2016 marked a historic milestone in QKD[110]. It is the world's first quantum-communication-dedicated satellite. Through Micius, long-distance satellite-to-ground QKD and entanglement distribution over ~ 1200 km were demonstrated. Later experiments[111] have also shown feasibility of QKD from microsatellites, raising hope for cost-effective satellite-based QKD networks. Recent reviews and studies[112, 32] outlined design choices for quantum payloads, ground station infrastructure, and highlights the emerging hybrid approaches to mitigate detection vulnerabilities.

While QKD is theoretically appealing, practical challenges remain in FSO SatComs[35]. Statistical fluctuations due to limited number of photons finite-key effects may limit the extractable secure key length[36]. Real hardware introduces deviations from the ideal protocol model. These imperfections can open side-channels or degrade security[35]. Most current QKD systems are point-to-point. Building a QKD network—with multiple users, routing, key management, possibly quantum repeaters or trusted nodes—is nontrivial. The design of QKD networks and integration with existing communication infrastructure remains an open research area[113].

3.3.4 Trust Root Security

Although there is a growing body of research on security for SatComs, most analyses remain agnostic to the underlying link technology and focus on link-level confidentiality, anti-jamming, or key agreement[9, 8]. By contrast, few recent works address trust root security in FSO SatCom system. Security issues in FSO SatComs spanning the physical, data link, and network layers, where indentifying trust management is one of the urgent open challenges.

This gap in the publicly available literature—a lack of detailed studies on how FSO SatCom architectures intersect with trust root security, supply chain assurance, key management infrastructure, and physical unclonable function—suggests a strong motivation for the present review. As optical links become more common in next-generation satellite constellations, the underlying spacecraft hardware and software stack will be just as critical a vector for attack as the communication channel itself[7, 15]. Without explicit attention to trust root security in the context of FSO SatCom, threats such as compromised optical terminal hardware, malicious spoofing attack, or weak identity management may remain unaddressed. Thereby undermining the security benefits that optical communications might promise. Highlighting and bridging this gap is not only timely but necessary for the secure adoption of FSO satellite systems.

3.4 Summary and Discussion

In this section, we have presented a comprehensive analysis of the safety and security challenges in RF and FSO SatComs. We categorize these challenges into four main domains: safety, physical layer security, protocol-level security, and trust root security, addressing both non-malicious and adversarial threats.

Safety and security research in RF SatComs is already quite comprehensive[4, 8, 14]. In contrast, FSO is often perceived as inherently secure, leading to relatively less focus on its security, particularly in the area of physical layer and trust root security[16, 126, 105].

From the safety perspective, we identify various types of interference that can degrade RF SatCom performance, such as multiple access interference (MAI), inter-beam interference (IBI), inter-satellite interference (ISI), and co-frequency interference between system[12]. These interference mechanisms are driven by the dense deployment of LEO constellations, spectrum scarcity, and the coexistence of terrestrial and satellite networks[81]. Safety issues in FSO SatComs primarily stem from the inherent characteristics of optical transmission, such as beam misalignment, atmospheric influences, and LoS obstructions[37]. These challenges require advanced tracking systems, site diversity, and adaptive transmission techniques to maintain reliable communication[16].

At the physical layer of RF SatComs, we focus on jamming and anti-jamming techniques, emphasizing the need for resilient methods such as frequency hopping, spread spectrum, advanced filtering techniques, and array antennas[9]. These methods are vital for maintaining signal availability and integrity[52]. FSO SatComs offer promising protection against traditional jamming and electromagnetic interference due to the narrow, directional nature of optical beams. However, optical jamming remains a viable threat, as adversaries can potentially disrupt the optical link by injecting interfering optical signals[127].

At the protocol-level, we explore RF security threats such as eavesdropping, replay/relay attacks, and spoofing. Given the broadcast nature of RF links, protecting the confidentiality and integrity of transmitted data is crucial. We discuss advanced countermeasures like robust key agreement protocols, authentication mechanisms[58, 59], and physical layer fingerprinting[55] to combat eavesdropping[10] and spoofing attacks. Effective replay and relay attack defenses are also emphasized, with particular focus on freshness checks, location binding, and cross-layer validation to ensure message authenticity[92]. The primary protocol-level security concern for FSO SatComs is eavesdropping. Mitigation techniques—authentication and key management protocols[34, 108], quantum key distribution (QKD)[109, 110, 32]—are discussed. Although QKD hold promise for enhancing protocol-level security, practical challenges such as photon loss, atmospheric effects, and hardware limitations remain significant hurdles that must be addressed in future research and development[35].

Trust root security, which underpins all other security layers, is identified as a key concern in modern SatCom systems. As satellites become increasingly integrated with terrestrial systems, ensuring the integrity of trust anchors—such as cryptographic keys and certificates—becomes critical[100]. We examine the challenges in key management, the importance of physical unclonable functions, and the vulnerabilities introduced by the deployments of COTS components in SatCom systems[101]. Secure lifecycle management of credentials and keys, along with hardware-based roots of trust, is essential to ensure the continued security of SatComs[66]. However, research specifically addressing trust root security in FSO systems is sparse, leaving potential vulnerabilities in the trustiness of both hardware and software components of FSO SatComs. Without proper attention to these trust root concerns, the security benefits of FSO communication could be undermined, leaving the entire satellite infrastructure vulnerable to spoofing attacks.

In conclusion, while SatComs continue to expand in capacity and complexity, integrating robust safety and security mechanisms in all layers will be critical to maintaining the reliability and trustworthiness of these systems. Further research is needed to explore more efficient and scalable solutions, particularly in the areas of FSO security. As SatComs converge with terrestrial systems, ensuring seamless safety and security integration between space and ground infrastructures will be essential for achieving truly reliable and secure global communication systems.

4 Learned Lessons and Future Directions

This section explores the evolution of safety and security in RF SatComs, and identifies future directions for FSO safety and security. Figure 7 spans multiple decades, from the early days of SatComs in the

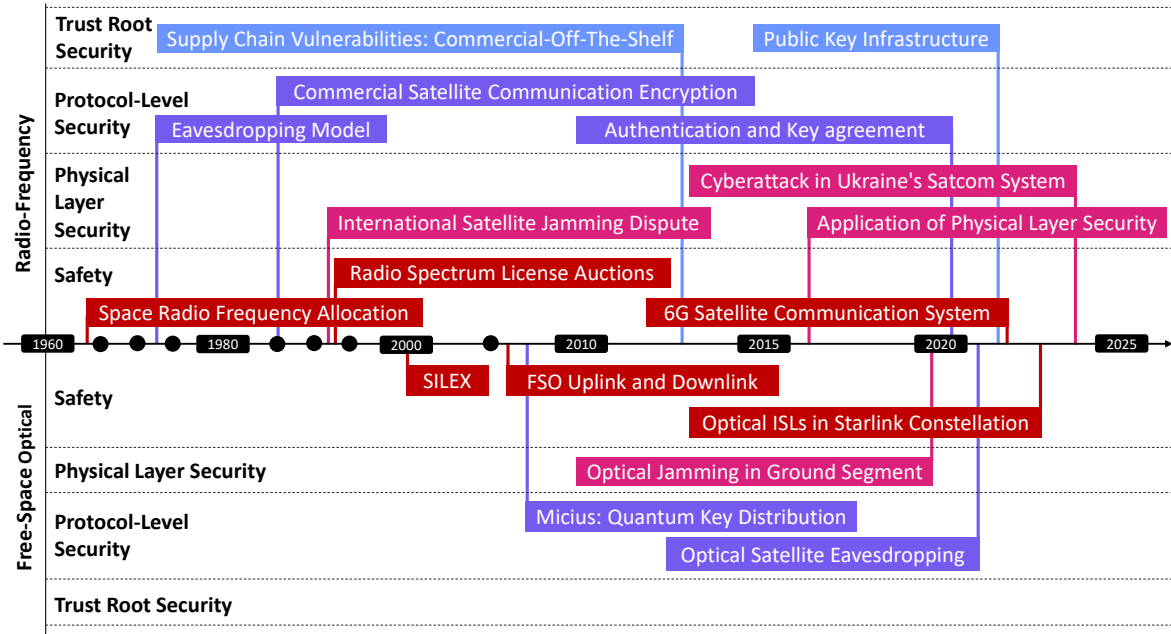


Figure 7. Safety and Security Evolution Timeline of SatComs

1960s to modern advances, helping contextualize the progression of safety and security in the evolving landscape of SatCom system.

We begin this section with a brief historical review of the key stages in RF SatCom security, highlighting important lessons learned. Then we outline potential advancements and research areas for improving safety and security of FSO SatComs.

4.1 Safety and Security Evolution of RF SatComs

The evolution of safety and security in SatComs has progressed through several stages, each responding to the growing demands and complexities of SatCom systems.

Early Development (1960s–1980s): Foundational Spectrum Safety and Basic Communication Security. In the early stages of RF SatComs, the focus was on the basic need for spectrum allocation and communications security. The administrative radio conference in 1963 [131] marked the first international effort to allocate dedicated frequency bands for SatComs. This recognition of spectrum as a limited resource laid the foundation for later discussions on secure and efficient communication. During this period, the primary security measures focused on signal scrambling and basic encryption techniques to prevent unauthorized interception of satellite broadcasts. Early TV SatCom broadcasts employed signal scrambling techniques to prevent unauthorized access [132]. These systems provided minimal but essential protection against unauthorized interception. The key lesson from this phase was the realization that even simple encryption mechanisms could provide essential protection against unauthorized access, which was crucial for securing early satellite services.

Expansion and Commercialization (1990s–2000s): Interference Safety and Jamming Threats. During the 1990s and 2000s, the commercialization of satellite services brought increased pressure on spectrum resources. In 1993, an international satellite jamming dispute between Indonesia and Tonga highlighted the growing concerns over spectrum contention and physical layer security [133]. Both countries threatened to jam each other's signals from their GEO satellite, marking the first recorded instance of a state using digital counterspace operations. Simultaneously, the concept of spectrum as a scarce and valuable resource gained prominence. Radio spectrum license auctions became a common approach since 1994, as national regulators moved away from traditional administrative methods toward more market-based spectrum allocation [134]. The increased competition for limited spectrum required

more sophisticated coexistence and interference management strategies to ensure reliable SatComs[12]. The commercialization of satellite services during this period highlighted the critical need for effective spectrum management and physical layer security development to protect SatComs from interference and jamming attacks.

Modern Era (2010s–Present): Multi Layer Security Architectures. In recent years, SatCom systems have been increasingly integrated with terrestrial networks, leading to more complex security requirements[135, 136]. The 2010s and beyond have seen a significant evolution in SatCom security, driven by the increasing complexity of constellations and the integration of satellites with terrestrial networks[137]. Physical layer security became a prominent focus during this period, with the early application of PLS methods in GEO and LEO satellite links to enforce secrecy constraints and protect against signal interception[138]. Simultaneously, protocol-level security evolved with advancements in authentication and key agreement protocols, essential for securing data integrity and preventing unauthorized access in satellite communications[30, 14]. Furthermore, the 2010s also saw the emergence of trust root security, addressing vulnerabilities in satellite supply chains, particularly due to the use of Commercial-Off-The-Shelf (COTS) components. The growing use of satellite systems, both commercial and governmental, expanded the potential attack surface, requiring more robust supply chain security measures[100, 103]. As satellite networks become increasingly interconnected, public key infrastructure has become essential for securing communications and verifying the trustworthiness of satellite systems and ground stations[68]. Together, these three pillars of security—physical layer security, protocol-level security, and trust root security—are now combined to create a comprehensive, system-wide security framework for SatComs, addressing the diverse threats faced by modern satellite systems.

4.2 Future Directions of FSO SatComs

The future of FSO SatCom holds great promise for enhancing global connectivity and overcoming current limitations of traditional RF-based systems. As FSO technology continues to evolve, several key areas are poised for further research and development. We explore these emerging directions, highlighting their potential impact on the future of FSO SatComs.

4.2.1 Open Source Datasets

One major challenge in FSO security research is the near absence of standardized, open source datasets that capture real physical layer phenomena under attack, such as eavesdropping, interference, or physical tampering. Unlike RF SatComs where community datasets and benchmarks are increasingly available, the FSO domain lacks publicly accessible experimental datasets that include channel measurements and physical layer effects in security contexts.

Most works on FSO security rely on analytical modeling and Monte-Carlo simulations of fading and eavesdropper channels rather than measured datasets[19, 37, 139]. This reliance on simulated data limits reproducibility, hinders the validation of security mechanisms against real optical attack behavior, and reduces opportunities for machine learning approaches that require large, labeled datasets.

The FSO SatCom community would benefit significantly from coordinated efforts to develop and release benchmark measurement datasets capturing diverse environmental conditions, pointing errors, interference patterns, and even controlled security attack scenarios. Such datasets could be invaluable for training and evaluating machine learning based mitigation algorithms in realistic settings, filling the important gap in the literature where the theoretical security benefits of FSO remain under-validated by real world data.

4.2.2 Machine Learning and Artificial Intelligence

With increasing complexity and dynamic behavior in FSO SatCom systems, integrating machine learning (ML) and artificial intelligence (AI) emerges as a key enabler for robustness, adaptability, and performance optimization. Recent research has demonstrated that AI techniques can effectively address core challenges in FSO satellite networks: notably atmospheric turbulence, pointing errors, link variability, and resource scheduling[140, 8].

In particular, ML based adaptive optics have been proposed to dynamically compensate wavefront distortions caused by atmospheric turbulence and platform motion, improving link stability and reducing bit-error rates without requiring manual tuning[141]. Meanwhile, deep learning methods such as convolutional neural networks (CNN) or generative models have been shown capable of mitigating turbulence induced distortions in FSO reception, thereby lowering symbol error rates and improving overall reliability[142, 139]. Beyond physical layer corrections, AI also enables efficient network-level and resource management functions in FSO SatComs. For instance, forecasting of channel availability can support proactive switching or fallback to hybrid RF/FSO links, enhancing overall availability[143]. More broadly, recent surveys[140] highlighted how ML and AI guided scheduling, beam-hopping, traffic forecasting, and resource allocation transform next-generation SatComs.

4.2.3 Relay-Assisted FSO Link

One of the most promising directions to overcome the inherent limitations of FSO links in satellite SatComs is the use of relay-assisted architectures. In particular, deploying intermediate nodes such as high-altitude platform stations (HAPS) or leveraging reconfigurable reflecting surfaces (RIS) can significantly improve coverage, link reliability, and resilience against blockages or turbulence.

HAPS operating in the stratosphere (e.g., at altitudes around 17–24 km) have been proposed as aerial relay nodes carrying FSO transceivers, enabling multi-hop optical links between ground stations or between ground and space[144, 124]. In a multi-hop HAP-based network, data from a ground transmitter is first sent optically to an FSO terminal on a HAP (uplink), then forwarded via inter-HAP FSO links among one or more HAPS, and finally downlinked optically from the last HAP to the destination on the ground[145]. This architecture helps to bypass ground obstacles and mitigate the adverse effects of atmospheric turbulence or weather that usually degrade terrestrial FSO performance. Theoretical analyses[144] showed that with careful design—including modeling of atmospheric attenuation, turbulence, pointing errors, and multi-hop attenuation—a ground-to-ground FSO link via HAP relays can achieve high data rate and reasonable reliability. Additionally, redundancy can be introduced via alternate slant links or multiple HAPS to improve survivability under link blockage due to Line-of-Sight (LoS) obstructions.

Deploying reconfigurable intelligent surfaces (RIS) is another emerging approach to improve FSO link performance, which steers, reflects or reshapes optical beams, thereby mitigating link impairments such as blockage, atmospheric turbulence, or pointing error. Recent studies[146] have extended RIS concept, originally developed for RF communications to FSO systems. In such RIS assisted FSO links, the RIS is composed of a large number of passive reflecting elements[147]. By appropriately configuring their phase and amplitude, the incident optical beam can be redirected towards the receiver even when LoS is blocked or degraded. Compared with HAP relays, RIS incur much lower cost and complexity, albeit typically enabling single-hop reflection rather than multi-hop relay across large distances.

4.2.4 Hybrid RF/FSO SatCom Systems

As the limitations of stand-alone RF or FSO links become more evident in practical SatCom deployments, hybrid RF/FSO systems—which combine the complementary strengths of RF and FSO communication—are emerging as a robust, flexible solution. In hybrid architectures, FSO links offer high throughput and spectral efficiency under favorable conditions, while RF links provide resilience to adverse weather, alignment jitter, or atmospheric turbulence. Consequently, hybrid RF/FSO is especially attractive for SatComs, where link conditions and environmental factors vary widely.

Recent literature classified hybrid RF/FSO systems into several categories depending on how the two links are used: hard switching, soft switching, relay-based, and diversity/combining schemes[148]. In hard-switching, the system dynamically selects either RF or FSO based on link quality or environmental conditions, using only one link at a time[149]. Soft-switching or link-combining schemes may merge signals from both RF and FSO paths for improved robustness and performance[150]. Relay-based hybrid architectures, often involving HAPS or RIS relays, enable dual-hop or multi-hop communication, combining FSO and RF segments as needed[151]. Such strategies are particularly promising for SatCom downlinks to ground when direct optical links are infeasible.

4.2.5 Routing for FSO Constellations

As constellations increasingly adopt FSO ISLs, efficient and robust routing strategies become critical[26]. Compared with terrestrial or RF-based networks, FSO constellations face unique challenges: rapid topology change due to orbital motion, limited link availability constrained by LoS, pointing errors, and tight resource constraints. Consequently, routing strategies must be tailored to these dynamics.

Recent literature has begun to propose routing strategies designed for FSO LEO constellations. A dual-layer architecture (MEO and LEO) with a regional division of the LEO layer was proposed to reduce complexity[152]. They applied a multi-objective reinforcement learning based routing algorithm to meet varied QoS requirements. A research[153] studied the routing problem in massive LEO constellations under distance-constrained inter-satellite communication. They propose a nearest-neighbor search algorithm to approximate globally optimal paths with low latency. The study also derived analytic bounds on latency relative to constellation parameters. Zhao et al. introduced a routing optimization framework that accounts for probabilistic ISL failures[154]. The method used a discrete time approach to generate static topology snapshots over orbital cycles and then employed an enhanced genetic algorithm to optimize routing under constraints such as task revenue, switching cost, and link failure probabilities.

4.2.6 Quantum Secure Direct Communication

Quantum secure direct communication (QSDC) is a quantum communication paradigm in which secret messages are directly transmitted over a quantum channel without the need for prior key agreement or subsequent classical encryption[155]. The appeal of QSDC lies in eliminating key management overhead and removing the classical ciphertext channel, potentially simplifying the communication stack and removing certain attack surfaces. In principle, a properly implemented QSDC protocol can achieve information-theoretic security, grounded in quantum mechanics[156]. Empirical demonstrations of QSDC on SatComs are not yet available, but the theoretical foundations and terrestrial experiments suggest potential for future satellite-based QSDC systems[157].

A landmark result[158] is the experimental implementation of free-space QSDC using single photons. In 2020, researchers reported a scheme using phase encoding and an asymmetric Mach–Zehnder interferometer, achieving a data transmission rate of 500 bits/s over 10 meters free-space. They also analyzed security under realistic attacks. Beyond discrete-variable (single-photon) experiments, continuous-variable QSDC has also seen experimental demonstration[159], indicating that QSDC can leverage different quantum-optical platforms. These experiments show the feasibility of QSDC. Building blocks are experimentally validated under FSO conditions.

Despite its promise, QSDC faces several nontrivial challenges before it can be applied in SatComs or large-scale network. Most experimental QSDC demonstrations remain over short distances. Extending to long free-space channels will face severe photon loss, turbulence, alignment issues, and atmospheric effects[157]. Real world implementations must deal with imperfect photon sources, detectors, channel noise, and potential side-channels[160]. Some advanced protocols attempt to mitigate this, but these often come with increased complexity or stricter requirements[155]. Therefore, while QSDC provides strong physical layer security in principle, there is significant work needed to bring it into practical SatComs.

5 Conclusion

The rapid expansion of SatCom systems, particularly with the integration of FSO technologies, underscores the need for a comprehensive understanding of both safety and security. While safety and security are often intertwined, it is crucial to differentiate between them to ensure effective risk management strategies. Our review highlights the unique characteristics of RF and FSO SatComs, focusing on their distinct challenges and vulnerabilities. By proposing a hierarchical taxonomy of safety and security issues, we provide a framework for understanding the specific risks associated with these communication technologies and their respective mitigation strategies.

Despite the common perception that FSO is inherently secure due to its narrow and highly directional beams, we reveal significant gaps in FSO security research. These gaps underscore the need for more focused investigations into the security aspects of FSO communication, particularly in the context of

potential intentional threats. Drawing from the lessons learned in the development of RF SatCom systems, we propose future directions to enhance the safety and security of FSO SatCom systems. Addressing these challenges will be essential for ensuring the continued reliability and resilience of SatCom systems in the face of evolving technological and security demands.

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Conflicts of interest

The authors declare no conflicts of interest.

Data availability statement

No data are associated with this article.

Author contribution statement

Weizhe Liu performed an extensive literature survey and drafted the manuscript. Shujiang Wu conceptualized the study and defined the overall structure and direction of the paper. Shutong Liu assisted with the preliminary literature review and prepared several figures. Qiang Wei and Yunhong Wang provided critical revisions and constructive suggestions that significantly improved the quality of the manuscript. All authors reviewed and approved the final version of the manuscript.

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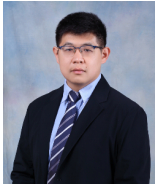
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